Nutanix Enterprise AI Guide

Nutanix Enterprise AI 2.4

August 11, 2025

Contents

ABOUT THIS PUBLICATION

This document provides information about Nutanix Enterprise AI.

Intended Audience

This document is intended for site personnel, the operation teams, and any tier or individual entity defined by you as equipped to perform site configurations and modifications, or anyone who intends to learn and develop an understanding of Nutanix Enterprise AI.

Related Documentation

The Nutanix Support Portal provides software download pages, documentation, compatibility, and other information.

Documentation	Description
Release Notes | Nutanix Enterprise AI	Release Notes for Nutanix Enterprise AI.
Nutanix Kubernetes Platform	This document provides information and code to enable the best-in-class hybrid cloud experience using the Nutanix Cloud Platform and Nutanix Kubernetes Platform.
Ports and Protocols	This page provides information about the ports that must be opened in the firewalls to enable Nutanix Enterprise AI to function.

GETTING STARTED WITH NUTANIX ENTERPRISE AI

This section provides an overview of Nutanix Enterprise AI that includes the following information:

- Architecture and a high-level end-to-end workflow
- Requirements and limitations
- Deployment and license types
- Installation and login procedure

Nutanix Enterprise AI Overview

Nutanix Enterprise AI is a comprehensive inference endpoint management product designed to streamline and optimize your AI model orchestration experience. Nutanix Enterprise AI allows you to select, deploy, and manage large language models (LLMs) on a Kubernetes® cluster.

The following diagram illustrates the high-level architecture of Nutanix Enterprise AI.

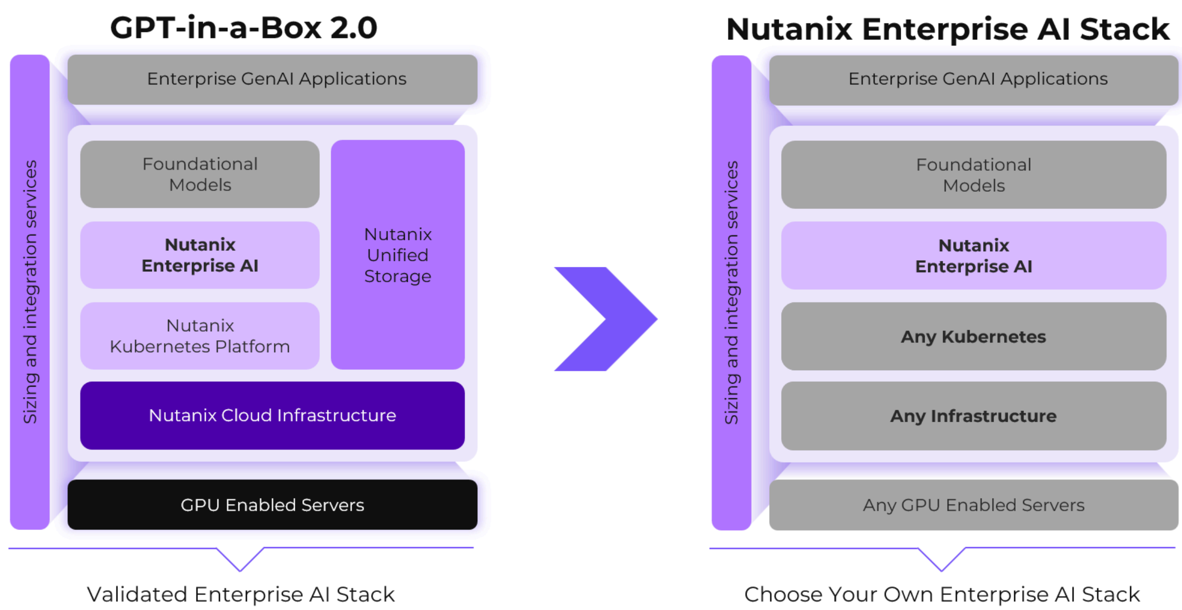

Figure 1: Nutanix Enterprise AI Architecture

Nutanix Enterprise AI comprises the following features:

Model access integration

Nutanix Enterprise AI supports selecting and deploying text-based generative AI LLMs from Hugging Face and NVIDIA.

Model access control

Nutanix Enterprise AI enables admin to control the models that can be deployed by the user.

Inference endpoint integration

Nutanix Enterprise AI supports creating an inference endpoint and sharing it with developers to incorporate into their AI applications. You can also manage and validate these endpoints to ensure that they are configured correctly.

API-based access control

Nutanix Enterprise AI supports accessing an inference endpoint by an application using API keys. API key management allows you to provide and revoke API access to ensure proper access security controls.

User access and role-based access control

Nutanix Enterprise AI supports role-based access control (RBAC), which you can configure to provide customized access permissions to users based on their assigned roles. The Users dashboard displays information about all the defined roles.

Enterprise user interface

Nutanix Enterprise AI has a simple and dynamic user interface that streamlines your deployment processes using one-click deployment.

Metrics dashboard

Nutanix Enterprise AI has a dynamic dashboard where you can monitor the health of your deployment with real-time monitoring tools that identify bottlenecks, track performance, and troubleshoot issues.

HTTP Proxy

Nutanix Enterprise AI provides support to configure forward proxy to download models from Hugging Face Model Hub.

Dark site deployment

Nutanix Enterprise AI provides support for deploying secure models in a dark site.

Built-in support for Nutanix products

Nutanix Enterprise AI supports seamless integration with existing Nutanix products, thus providing a robust and compliant solution.

You can deploy Nutanix Enterprise AI on Nutanix Kubernetes Platform (NKP), and public cloud Kubernetes platforms such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE).

NKP for Kubernetes v1.31 Kubernetes® is a registered trademark of The Linux Foundation in the United States and other countries and is used according to a license from The Linux Foundation.

Nutanix Enterprise AI Workflow

This section describes the high-level end-to-end workflow in Nutanix Enterprise AI.

The following procedure summarizes the workflow.

1. Install and log in to Nutanix Enterprise AI.
2. Import an LLM from Hugging Face, import an LLM manually, or import an NVIDIA NIM from NGC catalog.
3. Create an endpoint.
4. Test the endpoint.
5. Share the endpoint with your developer or data scientist user.

The following diagram illustrates the end-to-end workflow in Nutanix Enterprise AI.

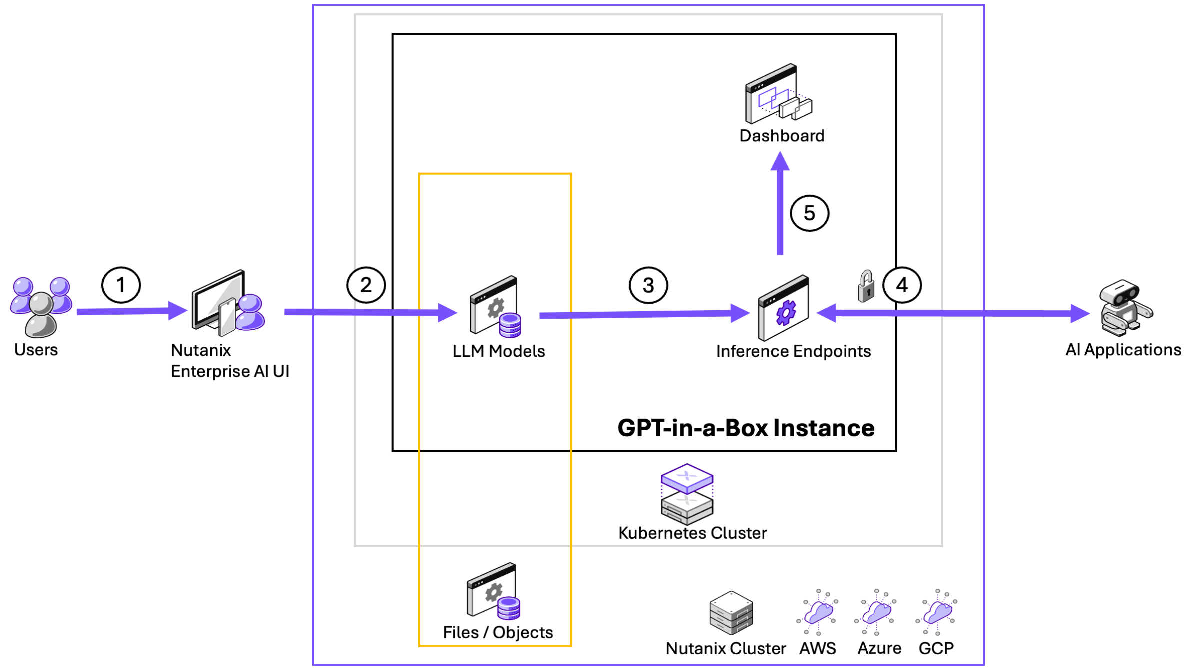

Figure 2: Nutanix Enterprise AI Workflow

Nutanix Enterprise AI Deployment Types

This section describes the types of Nutanix Enterprise AI deployments. The deployment types that you subscribe to determines which Nutanix Enterprise AI features are available to you.

Nutanix Enterprise AI supports distinct deployment methods to ensure the solution seamlessly integrates with your infrastructure and operational goals:

Nutanix Enterprise AI for GPT-in-a-Box 2.0

This deployment involves deploying Nutanix Enterprise AI Pro on a Nutanix validated solution stack, including Nutanix Cloud Infrastructure (NCI), Nutanix Kubernetes Platform (NKP), Nutanix Database Service (NDB), and Nutanix Unified Storage with 50TiB of Files, on your choice of compatible servers and hardware.

Nutanix Enterprise AI for Bare Metal

This deployment involves deploying Nutanix Enterprise AI Pro on bare metal servers, including hardware and GPU AI accelerators running on Nutanix Kubernetes Platform.

Nutanix Enterprise AI Standalone

This deployment involves deploying Nutanix Enterprise AI Pro on clusters running CNCF (Cloud Native Computing Foundation) compliant Kubernetes on bare metal servers, on-prem VMs, and public clouds such as Amazon EKS, Azure AKS, or Google GKE.

Nutanix Enterprise AI Licensing

Nutanix Enterprise AI is available with the Nutanix Enterprise AI Pro license.

The Nutanix Enterprise AI Pro license can be applied to all the deployment types of Nutanix Enterprise AI, and comprises the following features:

- Elegant, enterprise-grade user interface
- Choice of AI models (LLMs) from Hugging Face or NVIDIA NIM

- Upload your own AI models (LLMs)
- AI model pre-flight testing
- Partner API Token management for Hugging Face and NVIDIA NIM
- API Token creation and management
- API code samples
- Role-Based Access Controls (RBAC)
- AI model and API monitoring
- Kubernetes resource monitoring
- GPU usage monitoring
- Event auditing
- Integrated Nutanix Pulse reporting

Nutanix Enterprise AI includes a 45-day trial period. After the trial ends, you must purchase a valid license tier to continue using the service.

The licensing framework supports various deployment scenarios, including on-premises, hybrid, and cloud environments.

After your license expires, you have a 30-day grace period to renew or upgrade the license.

General Considerations

- The Nutanix Enterprise AI license consists of a vCPU and a GPU GB license. You must have a vCPU license to create an inference endpoint. A GPU GB license is optional and only required for GPU-based endpoints. Generate the licenses based on your specific requirements.

  > **Note:** You cannot purchase a GPU GB license without a vCPU license.

- During the trial period, if you add a valid license to your Nutanix Enterprise AI instance, the Nutanix Enterprise AI instance converts to a licensed version based on the resources defined in the applied license.
- When you add a vCPU license during the trial period, the GPU GB license trial period also ends, and the system updates the resources according to the applied license.
- You cannot create an inference endpoint if your vCPU license has expired, even if you have a valid GPU GB license. However, existing inference endpoints remain active. The system hibernates all endpoints after both the vCPU and GPU GB licenses expire. For more information, see Endpoint Hibernation on page 76.

Adding a License in Nutanix Enterprise AI

Add a new license using the Nutanix Enterprise AI user interface.

About this task

To add a new license, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Settings**.
   The **Third Party Credentials** page opens.

3. Click the **Licensing** tab.
The system displays the active trial license information.

4. Click **Add License**.
The system displays the **Add License** dialog box.

5. In the **License Key for vCPU** field, enter the vCPU license key that you generated from the Nutanix Licensing Portal.

   Generate a vCPU license key in any of the following ways:

   » Enter a vCPU license key generated from the Nutanix Licensing Portal using the Kubernetes cluster ID shown in the **Kubernetes Cluster ID** field.

   » Generate vCPU license key using a different Kubernetes cluster ID. Enter the vCPU license key and the corresponding Kubernetes cluster ID in the **Kubernetes Cluster ID** field.

   The meter type for the vCPU license key is vCPU.

6. (Optional) To add a GPT license, follow these steps:

   a. Select the **Using a NAI GPT Bundle** checkbox.

   b. In the **Prism ID** field, enter the Prism ID.

7. (Optional) In the **License Key for GPU (Optional)** field, enter the GPU GB license key that you generated from the Nutanix Licensing Portal.

   > **Note:**
   >
   > - Ensure that you enter the GPU GB license key generated from the Nutanix Licensing Portal using the Kubernetes cluster ID shown in the **Kubernetes Cluster ID** field.
   > - The meter type for the GPU GB license key is GB.

8. (Optional) To use existing license keys generated for another Nutanix Enterprise AI cluster, follow these steps:

   a. Select the **Keys generated using a different cluster ID** checkbox.
   The system displays the **Kubernetes Cluster ID** field.

   b. Enter the ID of the cluster used to generate the license keys from the Nutanix Licensing Portal.

9. Click **Add**.
The system applies the new license key and automatically adjusts the resources based on the added license.

Updating a License in Nutanix Enterprise AI

Update an existing license in Nutanix Enterprise AI.

About this task

To update an existing license, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.

2. From the left navigation pane, select **Settings**.
The **Third Party Credentials** page opens.

3. Click the **Licensing** tab.
   The system displays the active license information.

4. Click **Update License**.
   The system displays the **Update License** dialog box.

5. Make the required changes and click **Update**.
   The system updates the license details and automatically adjusts the resources based on the updated license.

Nutanix Enterprise AI Requirements

This section describes the requirements for deploying Nutanix Enterprise AI.

General Requirements

- For minimum software requirements to deploy Nutanix Enterprise AI in your environment , see Requirements in *Nutanix Enterprise AI Release Notes*.
- Ensure that you deploy a Kubernetes cluster with version 1.31.

  You can deploy Nutanix Enterprise AI on Nutanix Kubernetes Platform (NKP), and public cloud Kubernetes platforms such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE).
- Ensure that you configure a worker node pool with the necessary resources required to host an inference endpoint based on the inference engine and the required number of GPUs.

  For information on the Kubernetes platforms supported by the NVIDIA GPUs, see *Supported Operating Systems and Kubernetes Platforms* in *NVIDIA GPU Operator documentation*.

  For information on how to configure a worker node pool on an NKP cluster, see Configuring Node Pools in the *Nutanix Kubernetes Platform Guide*. For information on the minimum resources required for a worker node pool to host an inference endpoint, see Sizing Requirements on page 12.

  > **Note:** When you create an endpoint the system displays an error message if the worker node pool does not have the required number of GPUs available.
- After you configure a worker node pool, ensure that you taint the worker nodes with the key nvidia.com/gpu and the effect NoSchedule, to enable inference pods with matching toleration to be scheduled on the tainted worker node. Any pods that do not require GPUs will not be scheduled on the tainted worker node.
- Ensure that you deploy the necessary GPUs within a node pool in the Kubernetes cluster that hosts Nutanix Enterprise AI.

  For information on how to deploy GPUs on an NKP cluster, see Adding GPU Node Pool to a Nutanix Cluster in the *Nutanix Kubernetes Platform Guide*.
- Ensure that you install the NVIDIA GPU Operator to deploy LLMs using NVIDIA GPUs.

  For information on how to install the NVIDIA GPU Operator on an NKP cluster, see Configuring GPU for Kommander Clusters in the *Nutanix Kubernetes Platform Guide*. For information on how to install the NVIDIA

GPU Operator on a supported public cloud Kubernetes platform, see the *CSP Configurations* section in the NVIDIA GPU Operator documentation.

Nutanix recommends that you have one or more AI compatible GPUs deployed in your cluster. Nutanix Enterprise AI supports the following GPUs:

- NVIDIA Ada Lovelace L40S
- NVIDIA Hopper H100
- NVIDIA Hopper H100-NVL
- NVIDIA Ampere A100
- NVIDIA Hopper H200
- NVIDIA RTX PRO 6000 Blackwell Server Edition

For information on the Kubernetes platforms supported by the NVIDIA GPUs, see *Supported Operating Systems and Kubernetes Platforms* in *NVIDIA GPU Operator documentation*.

- Ensure that you create a ReadWriteMany (RWX) storage class that enables static or dynamic provisioning of NFS shares for persistent volumes. Configure the RWX storage class with `VolumeBindingMode` set to `Immediate`.

  For more information on how to create a storage class using Nutanix CSI driver, see Creating a Storage Class for Dynamic NFS Shares in the *CSI Volume Driver Guide*.

- Ensure that you configure the *LoadBalancer* service based on your Kubernetes cluster.

  For example, if you have an NKP cluster, ensure that you configure the MetalLB service in the cluster. For more information, see MetalLB in the *Nutanix Kubernetes Platform Guide*.

  The *LoadBalancer* service assigns an external IP address to the Envoy Ingress Gateway service in Nutanix Enterprise AI that external clients can use to connect to Nutanix Enterprise AI.

- Ensure that you configure a fully qualified domain name (FQDN) on the DNS domain that is accessible to your Kubernetes cluster using the external IP address of the Envoy Ingress Gateway service in Nutanix Enterprise AI.

  The FQDN is necessary to connect to Nutanix Enterprise AI.

- Ensure that you have a certificate authority (CA) signed TLS certificate for the configured FQDN.

  The TLS certificate is necessary to update the Envoy Ingress gateway after you install Nutanix Enterprise AI in the cluster.

  > **Note:** Nutanix Enterprise AI only supports HTTPS based connections with a valid TLS certificate.

Sizing Requirements

To deploy Nutanix Enterprise AI on a Kubernetes cluster in your environment, the cluster must have a minimum of three Kubernetes control plane nodes and three worker nodes. Additional worker nodes might be required for your environment and vary based on additional cluster workloads and high-availability requirements.

The following table lists the minimum compute and local storage requirements required to deploy Nutanix Enterprise AI on a Kubernetes Cluster.

> **Note:** To determine the cloud VM instance type based on the requirements mentioned in the following table, see the respective cloud service provider documentation.

Table 1: Minimum Compute and Storage Requirements for Deploying Nutanix Enterprise AI

Node Type	Number of Nodes (VM)	vCPU per Node	Memory per Node	Storage per Node	Total vCPU	Total Memory
Control Pane	3	4	16 GB	150 GB	12	48 GB
Worker	3	8	32 GB	150 GB	24	96 GB

The following table lists the minimum compute requirements for each Nutanix Enterprise AI platform component.

Table 2: Nutanix Enterprise AI Components - Minimum Compute Requirements

Component	App Name	Minimum vCPU Resources	Minimum Memory Resources	Purpose
API	nai-api	8	4 GiB	API Endpoint
Model Controller	nai-iep-model-controller	500m	500 MiB	LLM Model Downloader Operator
Frontend UI	nai-ui	2	2 GiB	Frontend
Database	nai-db	4	4 GiB	Application Data
Prometheus	prometheus-nai	8	4 GiB	Prometheus

The following table lists the minimum storage requirements for each Nutanix Enterprise AI platform component.

Table 3: Nutanix Enterprise AI Components - Minimum Storage Requirements

Component	App Name	Minimum Persistent Storage	Access Mode	Purpose
Database	nai-db	4 GiB	RWO	Application data stored in PostgreSQL database
Model	nai-api	20 GiB	RWO	Model custom resource used to initialize model storage
Prometheus	prometheus-nai	20 GiB	RWO	Prometheus metrics data stored in Prometheus database

Nutanix Enterprise AI uses various types of storage for the following purposes:

- To save the application and metrics data required to support the Nutanix Enterprise AI platform components.
- To download and store pre-validated LLMs required to optimize bootstrapping of LLM inferencing pods during deployment, scaling and upgrades.

- To share LLM files which enables high availability across multiple instances or replicas of LLM inferencing endpoints.
- To manually import downloaded pre-validated or custom LLM Models from existing NFS v4 Share or S3 Compatible Storage.

The following table lists the minimum storage sizing recommendation for NFS Storage used by RWX storageclass.

Table 4: Minimum Storage Sizing Recommendations

Component	Minimum Capacity	Storage Technology	Purpose
RWO (Block - RWO)	25 GiB	Block	Backend (Prometheus, PostgreSQL)
RWX (NFS - RWX)	2 TiB	NFS	Pre-validated or custom LLM models
S3 Compatible API	1 TiB	Object	Pre-validated or custom LLM models

GPU Requirements

The following table lists the pre-validated LLMs and the minimum number of supported GPUs required to deploy these LLMs.

> **Note:** The requirements mentioned in this table are applicable for entry-level deployments. Ensure that you plan your deployment based on your scale and traffic requirements.

Table 5: Minimum GPU Requirements

LLM Provider	LLM Name	GPU Models					
		NVIDIA L40S-48G	NVIDIA A100-80G	NVIDIA H100-80G	NVIDIA H100 NVL-94G	NVIDIA H 200-141G	NVIDIA RTX PRO 6000-96G
AI21 Labs	ai21labs/AI21-Jamba-1.5-Mini	4	2	2	2	1	Not supported
Google	google/gemma-2-9b-it	1	1	1	1	1	Not supported
	google/gemma-2-2b-it	1	1	1	1	1	Not supported
	google/vit-base-patch16-224	1	1	1	1	1	Not supported
IBM	ibm-granite/granite-embedding-107m-multilingual	1	1	1	1	1	Not supported
Meta	meta-llama/Llama-2-13b-chat-hf	1	1	1	1	1	Not supported

LLM Provider	LLM Name	GPU Models					
		NVIDIA L40S-48G	NVIDIA A100-80G	NVIDIA H100-80G	NVIDIA H100 NVL-94G	NVIDIA H 200-141G	NVIDIA RTX PRO 6000-96G
	meta-llama/ Llama-3.2-3b-Instruct	1	1	1	1	1	Not supported
	meta-llama/ Llama-3.2-1B-Instruct	1	1	1	1	1	Not supported
	meta-llama/Meta-Llama-3.1-8B-Instruct	1	1	1	1	1	Not supported
	meta-llama/Meta-Llama-3.1-70B-Instruct	4	2	2	2	2	Not supported
	meta-llama/ CodeLlama-7b-Instruct-hf	1	1	1	1	1	Not supported
	meta-llama/ CodeLlama-13b-Instruct-hf	1	1	1	1	1	Not supported
	meta-llama/ CodeLlama-34b-Instruct-hf	2	1	1	1	1	Not supported
	meta-llama/ CodeLlama-70b-Instruct-hf	4	2	2	2	2	Not supported
	meta-llama/ Llama-3.2-11B-Vision-Instruct	1	1	1	1	1	Not supported
	meta-llama/ Llama-3.2-90B-Vision-Instruct	4	4	4	4	2	Not supported
	meta-llama/ Llama-4-Scout-17B-16E	Not supported	Not supported	4	4	2	Not supported
	meta-llama/ Llama-Guard-3-8B	1	1	1	1	1	Not supported
	meta-llama/ Llama-3.3-70B-Instruct	4	2	2	2	2	Not supported
Mistral AI	mistralai/ Mistral-7B-Instruct-v0.3	1	1	1	1	1	Not supported

LLM Provider	LLM Name	GPU Models					
		NVIDIA L40S-48G	**NVIDIA A100-80G**	**NVIDIA H100-80G**	**NVIDIA H100 NVL-94G**	**NVIDIA H 200-141G**	**NVIDIA RTX PRO 6000-96G**
	mistralai/ Mixtral-8x7B-Instruct-v0.1	4	2	2	2	1	Not supported
	mistralai/ Mixtral-8x22B-Instruct-v0.1	Not supported	4	4	4	4	Not supported
	mistralai/ Mistral-Nemo-Instruct-2407	1	1	1	1	1	Not supported
NVIDIA	llama-3.1-8b-instruct	1	Not supported	Not supported	1	1	1
	llama-3.1-70b-instruct	4	Not supported	Not supported	2	1	Not supported
	llama-3.1-nemoguard-8b-content-safety	1	1	1	1	1	Not supported
	llama-3.2-nv-embedqa-1b-v2	1	1	1	1	1	Not supported
	llama-3.2-nv-rerankqa-1b-v2	1	1	1	1	1	Not supported
	llama-3.3-70b-instruct	4	Not supported	4	4	1	Not supported
	llama-3.3-nemotron-super-49b-v1	Not supported	Not supported	2	2	1	Not supported
	llama-3.1-nemoguard-8b-topic-control	1	Not supported	Not supported	1	1	Not supported
	llama-3.1-swallow-8b-instruct-v0.1	1	Not supported	Not supported	1	1	Not supported
	mixtral-8x7b-instruct-v0.1	4	Not supported	Not supported	2	1	Not supported
	mistral-7b-instruct-v0.3	1	Not supported	Not supported	1	1	Not supported
	phi-3-mini-4k-instruct	1	Not supported	Not supported	1	1	Not supported
	black-forest-labs/ flux.1-dev	Not supported	Not supported	Not supported	1	1	Not supported
	Mistral-nemo-12b-instruct	2	Not supported	Not supported	1	Not supported	Not supported

LLM Provider	LLM Name	GPU Models					
		NVIDIA L40S-48G	NVIDIA A100-80G	NVIDIA H100-80G	NVIDIA H100 NVL-94G	NVIDIA H 200-141G	NVIDIA RTX PRO 6000-96G
	Llama-3.2-90b-vision-instruct	Not supported	Not supported	Not supported	2	1	Not supported
	Llama-3.1-70b-instruct-pb24h2	Not supported	Not supported	Not supported	2	Not supported	Not supported
	Llama-3.1-nemotron-70b-instruct	Not supported	Not supported	Not supported	2	Not supported	Not supported
	Llama-3.1-8b-instruct-pb24h2	1	Not supported	Not supported	1	1	Not supported
	gpt-oss-20b	Not supported	Not supported	1	1	Not supported	Not supported
	gpt-oss-120b	Not supported	Not supported	2	1	Not supported	Not supported
Unsloth	unsloth/ Llama-3.3-70B-Instruct-bnb-4bit	1	1	1	1	1	Not supported
Facebook	facebook/deit-base-distilled-patch16-224	1	1	1	1	1	Not supported
Stability AI	stable-diffusion-v1-5/stable-diffusion-v1-5	1	1	1	1	1	Not supported
Cross-Encoder	cross-encoder/ ms-marco-MiniLM-L6-v2	1	1	1	1	1	Not supported

The following table lists the minimum resources required to host an inference endpoint based on the inference engine and the number of GPUs.

> **Note:**
>
> - The minimum worker node pool size must be defined based on the sum of the CPU and memory resources required for all running inference workloads.
> - The requirements mentioned in this table are suggestive. Deploying larger LLMs requires more resources.

Table 6: Minimum GPU Node Pool Requirements

Type of Inference Engine	Resource Requirement	Number of GPUs			
		1	2	4	8
NVIDIA NIM	CPU (number of cores)	12	12	12	12
	Memory (GiB)	32	32	32	32
TGI	CPU (number of cores)	4	8	16	32
	Memory (GiB)	12	24	48	96
vLLM	CPU (number of cores)	8	8	8	8
	Memory (GiB)	16	32	64	128

The following table lists the virtual machine types supported by the GPU model necessary to deploy Nutanix Enterprise AI in public cloud Kubernetes platforms.

> **Note:** For the latest virtual machine types supported by a GPU model, see the respective cloud service provider documentation.

Table 7: GPU Supported Virtual Machine Type

Cloud Service Provider	Platform Name	GPU Supported Virtual Machine Type	GPU Model
AWS	Amazon Elastic Kubernetes Service (EKS)	EC2 G6e	NVIDIA Ada Lovelace L40S
		EC2 P5	NVIDIA Hopper H100
		EC2 P4	NVIDIA Ampere A100
		EC2 P5e	NVIDIA Hopper H 200
GCP	Google Kubernetes Engine (GKE)	A3 VM	NVIDIA Hopper H100
		A2 VM	NVIDIA Ampere A100
Azure	Azure Kubernetes Service (AKS)	NCads_H100_v5-series	NVIDIA Hopper H100
		NCCads_H100_v5-series	
		NC_A100_v4-series	NVIDIA Ampere A100

Nutanix Enterprise AI Limitations

The following limitations apply when you deploy Nutanix Enterprise AI at your site.

- Nutanix Enterprise AI supports Hugging Face and NVIDIA NIM formatted LLMs only.
- Nutanix Enterprise AI supports local users and permissions management. It does not support Active Directory integration.
- Nutanix Enterprise AI supports GPU operator configured with GPU Passthrough only.

- Nutanix Enterprise AI supports x86 architecture only.

Kubernetes Cluster Setup

This section provides information on setting up Kubernetes clusters for Nutanix Enterprise AI.

You can deploy Nutanix Enterprise AI on Nutanix Kubernetes Platform (NKP), and public cloud Kubernetes platforms such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE).

Setting up a Nutanix Kubernetes Platform Cluster

Set up your Nutanix Kubernetes Platform (NKP) cluster for Nutanix Enterprise AI.

About this task

To set up your NKP cluster before you deploy Nutanix Enterprise AI, follow these steps:

Procedure

1. Create an NKP cluster.

   Ensure that you use Kubernetes version 1.31 and the nodes are Ubuntu based images. For more information, see the Nutanix Kubernetes® Platform Guide.

2. Add NKP Pro or Ultimate license to your NKP Cluster.

   For more information, see Add an NKP License in the *Nutanix Kubernetes® Platform Guide*.

3. Ensure that your NKP cluster meets all the requirements mentioned in Nutanix Enterprise AI Requirements on page 11.

4. Create a Nutanix Files server with sufficient storage required to save the models.

   For more information, see Creating a File Server in the *Nutanix Files User Guide*.

   For information on the size of the models, see Table 18: Pre-validated Models on page 51.

5. Create a Network File System (NFS v4) export in the Nutanix Files server to import a custom model or to import a model manually.

   For more information, see Creating an NFS Export in the *Nutanix Files User Guide*.

6. Install the following components in your NKP cluster:

   - Cert-manager v1.16.4
   - Envoy Gateway v1.3.2
   - Kserve: v0.15.0
   - NVIDIA GPU Operator: v25.3.0
   - Kube-prometheus-stack: 70.4.2

   Note:

   - You can install Nutanix Enterprise AI on an NKP cluster even if the NKP cluster has a version of kube-prometheus-stack that is lower than the minimum version of kube-prometheus-stack supported by Nutanix Enterprise AI. There is no need to upgrade to the minimum version of kube-prometheus-stack supported by Nutanix Enterprise AI.

- If you have **Prometheus Monitoring** enabled in an NKP management cluster or an NKP managed cluster, do not install kube-prometheus-stack in the cluster.
- For the metrics in Nutanix Enterprise AI to function as expected:
  - Ensure that you enable prometheus-node-exporter when you install kube-prometheus-stack.
  - If you enable kube-rbac-proxy with prometheus-node-exporter, configure the bearerTokenFile filepath and set the metrics endpoint to use HTTPS for the node exporter service monitor in the `values.yaml` file. Also, add the following role permissions to the NAI Prometheus service account:

```
rules:
            - apiGroups: [""]
              resources: ["services/prometheus-node-exporter"]
              verbs:
                - get
```

  - Ensure that the labels for the service monitors in the nai-system namespace match the respective service labels installed in the NKP cluster.

- Nutanix CSI Driver: 3.3

Note: By default, a Nutanix Kubernetes Platform (NKP) cluster with an *NKP Pro* license has cert-manager, Nutanix CSI Storage, and Nutanix CSI Snapshot installed. For more information, see the Nutanix Kubernetes® Platform Guide.

What to do next

Complete the steps provided in Installing or Upgrading Nutanix Enterprise AI on page 26.

Setting up an Amazon Elastic Kubernetes Service Cluster

Set up your Amazon Elastic Kubernetes Service (EKS) cluster for Nutanix Enterprise AI.

Before you begin

Ensure that your cluster meets all the requirements mentioned in Nutanix Enterprise AI Requirements on page 11.

About this task

To set up your EKS cluster before you deploy Nutanix Enterprise AI, follow these steps:

Warning: This procedure is not applicable for Nutanix Kubernetes Platform (NKP) managed or attached EKS clusters.

Procedure

1. Configure the AWS CLI.

   For more information, see AWS documentation.

2. Configure an Amazon virtual private cloud (VPC) with public and private subnets for EKS using CloudFormation.

   For more information, see AWS documentation.

3. Create an Amazon EKS cluster.

   For more information, see AWS documentation.

4. Create a kubeconfig file to connect kubectl to the EKS cluster.

   For more information, see AWS documentation.

5. Create an EKS role for the EKS cluster.

   For more information, see AWS documentation.

6. Create an Elastic Compute Cloud (EC2) role for the EKS cluster.

   For more information, see AWS documentation.

7. Create a default node group. Make sure the nodes have accelerator AVX2 or newer.

   The recommended instance type is from instance family C5.

8. Create a GPU node group on the EKS Cluster.

   For more information, see AWS documentation.

9. Configure the Amazon EBS CSI driver and the RWO storageclass.

   For more information, see AWS documentation.

10. Configure the Amazon EFS CSI driver and the RWX storageclass.

    For more information, see AWS documentation.

11. Configure the NAI-Core helm chart `values.yaml` file with the following parameters:

```
# Storage class name to be used by nai-db and prometheus db
defaultStorageClassName: gp2

# nai-monitoring stack values for nai-monitoring stack deployment in AWS
 environment
naiMonitoring:
  ## Component scraping node exporter
  ##
  nodeExporter:
    serviceMonitor:
      enabled: true
      endpoint:
        port: http-metrics
        scheme: http
        targetPort: 9100
        scheme: http
      namespaceSelector:
        matchNames:
        - prometheus
      serviceSelector:
        matchLabels:
          app.kubernetes.io/name: prometheus-node-exporter
          app.kubernetes.io/component: metrics

  ## Component scraping dcgm exporter
  ##
  dcgmExporter:
    podLevelMetrics: true
    serviceMonitor:
      enabled: true
      endpoint:
        targetPort: 9400
      namespaceSelector:
```

```
        matchNames:
        - gpu-operator
      serviceSelector:
        matchLabels:
          app: nvidia-dcgm-exporter
```

12. Install the following components in your EKS cluster:

- Envoy Gateway v1.3.2
- Cert-manager: v1.16.4
- Kserve: 0.15.0
- NVIDIA GPU Operator: v25.3.0
- Kube-prometheus-stack: 70.4.2

> **Note:** For the metrics in Nutanix Enterprise AI to function as expected, ensure that the labels for the service monitors in the nai-system namespace match the respective service labels installed in the EKS cluster.

What to do next

Complete the steps provided in Installing or Upgrading Nutanix Enterprise AI on page 26.

Setting up an Azure Kubernetes Service Cluster

Set up your Azure Kubernetes Service (AKS) cluster for Nutanix Enterprise AI.

Before you begin

Ensure that your cluster meets all the requirements mentioned in Nutanix Enterprise AI Requirements on page 11.

About this task

To set up your AKS cluster before you deploy Nutanix Enterprise AI, follow these steps:

> **Warning:** This procedure is not applicable for Nutanix Kubernetes Platform (NKP) managed or attached AKS clusters.

Procedure

1. Configure Azure CLI.

For more information, see Azure documentation.

2. Create an Azure resource group.

For more information, see Azure documentation.

3. Create an AKS cluster.

For more information, see Azure documentation.

4. Configure the AKS preview extension.

For more information, see Azure documentation.

5. Create a default node group. Make sure the nodes have accelerator AVX2 or newer.

6. Add a GPU node pool to the AKS cluster using Azure CLI.

   For more information, see Azure documentation.

   > **Note:** Ensure that you skip installing the GPU driver when adding the GPU node pool to the cluster. For more information, see Azure documentation.

7. Configure the *NAI-Core* helm chart *values.yaml* file with the following parameters.

```
$ cat <<EOF >> aks-values.yaml
naiMonitoring:
  nodeExporter:
    serviceMonitor:
      enabled: true
      endpoint:
        port: http-metrics
        scheme: http
        targetPort: 9100
        scheme: http
      namespaceSelector:
        matchNames:
        - kube-prometheus-stack-namespace
      serviceSelector:
        matchLabels:
          app.kubernetes.io/name: prometheus-node-exporter
          app.kubernetes.io/component: metrics
          app.kubernetes.io/version: 1.8.1

  dcgmExporter:
    podLevelMetrics: true
    serviceMonitor:
      enabled: true
      endpoint:
        targetPort: 9400
      namespaceSelector:
        matchNames:
        - gpu-operator-namespace
      serviceSelector:
        matchLabels:
          app: nvidia-dcgm-exporter
EOF
```

   Where:

   - Replace `kube-prometheus-stack-namespace` with the namespace used to install kube-prometheus-stack. For example, `prometheus`.
   - Replace `gpu-operator-namespace` with the namespace used to install the NVIDIA GPU operator. For example, `gpu-operator`.

8. Install the following components in your AKS cluster:

    - Envoy Gateway v1.3.2
    - Cert-manager: v1.16.4
    - Kserve: 0.15.0
    - NVIDIA GPU Operator: v25.3.0
    - Kube-prometheus-stack: 70.4.2

    > **Note:** For the metrics in Nutanix Enterprise AI to function as expected, ensure that the labels for the service monitors in the nai-system namespace match the respective service labels installed in the AKS cluster.

What to do next

Complete the steps provided in Installing or Upgrading Nutanix Enterprise AI on page 26.

Setting up a Google Kubernetes Engine Cluster

Set up your Google Kubernetes Engine (GKE) cluster for Nutanix Enterprise AI.

Before you begin

Ensure that your cluster meets all the requirements mentioned in Nutanix Enterprise AI Requirements on page 11.

About this task

To set up your GKE cluster before you deploy Nutanix Enterprise AI, follow these steps:

> **Warning:** This procedure is not applicable for Nutanix Kubernetes Platform (NKP) managed or attached GKE clusters.

Procedure

1. Configure and authorize the Google Cloud CLI.

    For more information, see Google Cloud documentation.

2. Enable *Required Services*.

    This is required to enable Google files CSI driver and default RWX storage classes. For more information, see Google Cloud documentation.

3. Create a GKE cluster based on the *GKE GPU Operator NVIDIA Driver Manager* workflow.

    For more information, see Google Cloud documentation.

4. Create a default node group. Make sure the nodes have accelerator AVX2 or newer.

5. Create a GKE node pool and add it to the cluster.

    For more information, see Google Cloud documentation.

    > **Note:** Ensure that you skip installing the Google GPU driver when adding the GKE node pool to the cluster.

6. Configure the node pool to create the GPU operator namespace and deploy the GPU operator resource quota.

    For more information, see Google Cloud documentation.

7. Configure the *NAI-Core* helm chart *values.yaml* file with the following parameters.

```
$ cat <<EOF >> gke-values.yaml
naiMonitoring:
  nodeExporter:
    serviceMonitor:
      enabled: true
      endpoint:
        port: http-metrics
        scheme: http
        targetPort: 9100
        scheme: http
      namespaceSelector:
        matchNames:
        - kube-prometheus-stack-namespace
      serviceSelector:
        matchLabels:
          app.kubernetes.io/name: prometheus-node-exporter
          app.kubernetes.io/component: metrics
          app.kubernetes.io/version: 1.8.1

  dcgmExporter:
    podLevelMetrics: true
    serviceMonitor:
      enabled: true
      endpoint:
        targetPort: 9400
      namespaceSelector:
        matchNames:
        - gpu-operator-namespace
      serviceSelector:
        matchLabels:
          app: nvidia-dcgm-exporter
EOF
```

Where:

- Replace `kube-prometheus-stack-namespace` with the namespace used to install kube-prometheus-stack. For example, `prometheus`.
- Replace `gpu-operator-namespace` with the namespace used to install the NVIDIA GPU operator. For example, `gpu-operator`.

8. Install the following components in your GKE cluster:

- Envoy Gateway v1.3.2
- Cert-manager: v1.16.4
- Kserve: 0.15.0
- NVIDIA GPU Operator: v25.3.0
- Kube-prometheus-stack: 70.4.2

> **Note:** For the metrics in Nutanix Enterprise AI to function as expected, ensure that the labels for the service monitors in the nai-system namespace match the respective service labels installed in the GKE cluster.

What to do next

Complete the steps provided in Installing or Upgrading Nutanix Enterprise AI on page 26.

Installing or Upgrading Nutanix Enterprise AI

Install or upgrade Nutanix Enterprise AI on a Kubernetes cluster using Helm charts and the Docker Hub container image library.

Before you begin

> **Note:** If you installed the existing Nutanix Enterprise AI version using the NKP catalog, then upgrade it using the NKP catalog only. For information on how to upgrade Nutanix Enterprise AI using the NKP catalog, see Nutanix Enterprise AI in Workspace in the *Nutanix Kubernetes® Platform Guide*.

- Ensure that your site meets all the requirements mentioned in Nutanix Enterprise AI Requirements on page 11.
- Ensure that you set up your cluster based on the requirements mentioned in Kubernetes Cluster Setup on page 19.
- Log in to the Nutanix Support portal and generate a Docker Hub access token. For more information, see Generating Nutanix Docker Hub Access Tokens.

  The access token is required to obtain the Helm chart and the Nutanix Enterprise AI container artifacts from the Nutanix Enterprise AI page.
- Install *Kubectl* and *Helm CLI*.
- To make any additional configuration changes to the *nai-core* helm chart, see Nutanix Enterprise AI Configuration Parameters on page 28.
- Admin must perform the following one-time operation on upgrade:
  1. Scan the audit logs for the following **Description**: `Updated state of <endpoint-name> endpoint from active to hibernated`

     For example, `Updated state of llamavllm endpoint from active to hibernated`
  2. Activate the corresponding endpoints.

     For more information, see Resuming an Endpoint.

About this task

To install or upgrade Nutanix Enterprise AI on a Kubernetes cluster, follow these steps:

Procedure

1. Add and update the Nutanix helm repository, which contains the *nai-core* helm chart:

```
$ helm repo add ntnx-charts https://nutanix.github.io/helm-releases && helm repo
 update ntnx-charts
```

2. Search for the version of the *nai-core* helm chart available for installation in the Nutanix helm repository:

```
$ helm search repo ntnx-charts/nai-core --versions
```

3. Pull the *nai-core* helm chart and extract it to your local repository:

```
$ helm pull ntnx-charts/nai-core --version=nai-core-version --untar=true
```

   Replace `nai-core-version` with the version of *nai-core* helm chart that you obtained in Step 2.

4. Install or upgrade the *nai-core* helm chart in the *nai-system* namespace:

```
$ helm upgrade --install nai-core ntnx-charts/nai-core --version=NAI_CORE_VERSION -n
 nai-system --create-namespace --wait \
```

```
        --set imagePullSecret.credentials.username=DOCKER_USERNAME \
        --set imagePullSecret.credentials.email=DOCKER_USERNAME \
        --set imagePullSecret.credentials.password=DOCKER_API_KEY \
        --insecure-skip-tls-verify \
        --set naiApi.storageClassName=NAI_API_RWX_STORAGECLASS \
        --set defaultStorageClassName=NAI_DEFAULT_RWO_STORAGECLASS \
        -f ./nai-core/values.yaml
```

Replace the following values:

- `NAI_CORE_VERSION` with the version of *nai-core* helm chart that you obtained in Step 2.
- `DOCKER_USERNAME` with the username of your Docker account.
- `DOCKER_API_KEY` with the API key or the access token used to access your Docker account.
- `NAI_API_RWX_STORAGECLASS` with the name of the ReadWriteMany (RWX) storage class.

  > **Note:** If you created a dedicated RWX storage class, provide the name as `nai-nfs-storage`. If you did not create a dedicated RWX storage class, provide the name used in `naiApi.storageClassName`.

- `NAI_DEFAULT_RWO_STORAGECLASS` with the name of the ReadWriteOnce (RWO) storage class.
- `values.yaml` with the following templates:
  - EKS clusters: `aws-values.yaml` template available in the *nai-core* helm chart.
  - NKP clusters: `nkp-values.yaml` template available in the *nai-core* helm chart.
  - AKS clusters: `aks-values.yaml` template created in Step 6 of Setting up an Azure Kubernetes Service Cluster on page 22.
  - GKE clusters: `gke-values.yaml` template created in Step 6 of Setting up a Google Kubernetes Engine Cluster on page 24.

What to do next

- After you upgrade your Nutanix Enterprise AI instance to version 2.3 or later, hibernate and resume any existing CPU-based endpoint to ensure that the endpoint uses the latest runtime with Intel extensions for inferencing. If the CPU generations differ between your production and pre-production environments, your validation results might vary.
- Ensure that all the pods are successfully deployed in the nai-system namespace and are in the ready state by:

  ```
  $ kubectl get pods -n nai-system
  ```

  The following is a sample output of the command:

  ```
  NAME                                            READY    STATUS       RESTARTS    AGE
  nai-api-6c85c95866-bkwm2                                  1/1       Running      0
   23m
  nai-api-db-migrate-h5sqd-xxhsr                            0/1       Completed    0
   23m
  nai-db-0                                                  1/1       Running      0
   23m
  nai-iep-model-controller-85df96896d-2zzrf                 1/1       Running      0
   23m
  nai-ui-77f6b4ddc6-nhk4c                                   1/1       Running      0
   23m
  prometheus-nai-0                                          2/2       Running      0
   23m
  ```

Nutanix Enterprise AI Configuration Parameters

This section provides information about the configurable parameters of the *NAI-Core* helm chart.

The following tables list the configurable parameters of the *NAI-Core* helm chart and their default values.

Table 8: Common Parameters

Key	Description	Default Value
labels.app.kubernetes.io/name	Labels common to add components of nai-core	nai-iep

Table 9: Image Pull Secret Parameters

Key	Description	Default Value
labels.app.kubernetes.io/name	Labels common to add components of nai-core	nai-iep
imagePullSecret.name	Name of the image pull secret	nai-iep-secret
imagePullSecret.credentials.registry	Image registry credentials	https://index.docker.io/v1/
imagePullSecret.credentials.username	Image registry username	USERNAME
imagePullSecret.credentials.password	Image registry password	PASSWORD
imagePullSecret.credentials.email	Image registry email	EMAIL
defaultStorageClassName	Storage class name to be used by nai-db and prometheus db	

Table 10: Storage Parameters

Key	Description	Default Value
defaultStorageClassName	Storage class name to be used by nai-db and prometheus db	

Table 11: NAI IEP Operator Parameters

Key	Description	Default Value
naiIepOperator.iepOperatorImage.image	IEP operator image name	docker.io/nutanix/nai-iep-operator
naiIepOperator.iepOperatorImage.tag	IEP operator image tag	latest
naiIepOperator.iepOperatorResources.limits.cpu	IEP operator CPU limits	500m
naiIepOperator.iepOperatorResources.limits.memory	IEP operator memory limits	500Mi
naiIepOperator.iepOperatorResources.requests.cpu	IEP operator CPU requests	100m
naiIepOperator.iepOperatorResources.requests.memory	IEP operator memory requests	100Mi

Key	Description	Default Value
naiIepOperator.modelProcessorImage.image	Model Processor image name	docker.io/nutanix/nai-model-processor
naiIepOperator.modelProcessorImage.tag	Model Processor image tag	latest
naiIepOperator.modelProcessorResources.limits.cpu	Model Processor CPU limits	
naiIepOperator.modelProcessorResources.limits.memory	Model Processor memory limits	
naiIepOperator.modelProcessorResources.requests.cpu	Model Processor CPU requests	
naiIepOperator.modelProcessorResources.requests.memory	Model Processor memory requests	
naiIepOperator.retainModelProcessorJob	Determines whether the model processor job should be retained after completion.	false

Table 12: NAI Inference UI Parameters

Key	Description	Default Value
naiInferenceUi.naiUiImage.image	The name of the NAI UI Docker image	docker.io/nutanix/nai-inference-ui
naiInferenceUi.naiUiImage.tag	The tag of the Docker image to be used	latest
naiInferenceUi.resources.requests.cpu	NAI UI container CPU requests	1
naiInferenceUi.resources.requests.memory	NAI UI container memory requests	1Gi
naiInferenceUi.resources.limits.cpu	NAI UI container CPU limits	2
naiInferenceUi.resources.limits.memory	NAI UI container memory limits	2Gi
naiInferenceUi.naiApiUrl	The URL for the NAI API service	http://nai-api.nai-system.svc.cluster.local:7000

Table 13: NAI API Parameters

Key	Description	Default Value
naiApi.naiApiImage.image	The name of the NAI API Docker image	docker.io/nutanix/nai-api
naiApi.naiApiImage.tag	The tag of the Docker image to be used	latest
naiApi.naiApiResources.requests.cpu	NAI API container CPU requests	2
naiApi.naiApiResources.requests.memory	NAI API container memory requests	2Gi
naiApi.naiApiResources.limits.cpu	NAI API container CPU limits	8
naiApi.naiApiResources.limits.memory	NAI API container memory limits	4Gi

Key	Description	Default Value
naiApi.naiMigrateInitContainerResources.requests.cpu	NAI API migrate init container CPU requests	100m
naiApi.naiMigrateInitContainerResources.requests.memory	NAI API migrate init container memory requests	50Mi
naiApi.naiMigrateInitContainerResources.limits.cpu	NAI API migrate init container CPU limits	1
naiApi.naiMigrateInitContainerResources.limits.memory	NAI API migrate init container memory limits	1Gi
naiApi.naiMigrateJobResources.requests.cpu	NAI API migrate job container CPU requests	1
naiApi.naiMigrateJobResources.requests.memory	NAI API migrate job container memory requests	1Gi
naiApi.naiMigrateJobResources.limits.cpu	NAI API migrate job container CPU limits	1
naiApi.naiMigrateJobResources.limits.memory	NAI API migrate job container memory limits	1Gi
naiApi.superAdmin.username	Super admin username	admin
naiApi.superAdmin.password	Super admin password	Nutanix.123
naiApi.superAdmin.email	Super admin email	admin@nutanix.com
naiApi.superAdmin.firstName	Super admin first name	admin
naiApi.storageClassName	Storage class name to be used nai-iep for storing models	nai-nfs-storage
naiApi.supportedKserveRuntimeImage	Supported Kserve Runtime Image in model catalog	docker.io/nutanix/nai-kserve-huggingfaceserver:v0.14.0
naiApi.supportedTGIImage	Supported TGI Runtime Image	docker.io/nutanix/nai-tgi
naiApi.supportedTGIImageTag	Supported TGI Runtime Image tag	2.3.1-825f39d
naiApi.naiMonitoringUrl	The URL for NAI prometheus service	http://nai-prometheus.nai-system.svc.cluster.local:9090

Table 14: NAI Database Parameters

Key	Description	Default Value
naiDatabase.naiDbImage.image	The name of the NAI DB Docker image	docker.io/nutanix/nai-postgres:16.1-alpine
naiDatabase.service.port	Postgres sql service port	5432
naiDatabase.postgresql.postgresUser	Postgres sql user	nai-api-user
naiDatabase.postgresql.postgresPassword	Postgres sql password	nai-api-password
naiDatabase.postgresql.postgresDatabase	Postgres sql database	nai_iep
naiDatabase.postgresql.postgresPort	Postgres sql port	5432

Key	Description	Default Value
naiDatabase.postgresql.postgresHost	Postgres sql host	nai-db
naiDatabase.resources.requests.cpu	NAI DB container CPU requests	2
naiDatabase.resources.requests.memory	NAI DB container memory requests	2Gi
naiDatabase.resources.limits.cpu	NAI DB container CPU limits	4
naiDatabase.resources.limits.memory	NAI DB container memory limits	4Gi
naiDatabase.storageSpec.resources.requests.storage	NAI DB storage requests	4Gi

Table 15: NAI Monitoring Parameters

Key	Description	Default Value
naiMonitoring.prometheus.image.registry	Prometheus image registry	quay.io
naiMonitoring.prometheus.image.repository	Prometheus image repository	prometheus/prometheus
naiMonitoring.prometheus.image.tag	Prometheus image tag	v2.54.0
naiMonitoring.prometheus.version	Prometheus version	v2.54.0
naiMonitoring.prometheus.replicas	Number of prometheus pods to deploy	1
naiMonitoring.prometheus.resources.requests.cpu	Prometheus pod CPU requests	2
naiMonitoring.prometheus.resources.requests.memory	Prometheus pod memory requests	2Gi
naiMonitoring.prometheus.resources.limits.cpu	Prometheus pod CPU limits	8
naiMonitoring.prometheus.resources.limits.memory	Prometheus pod memory limits	4Gi
naiMonitoring.prometheus.storageSpec.resources.requests.storage	Prometheus storage requests	20Gi
naiMonitoring.prometheus.persistentVolumeClaimRetentionPolicy.whenScaled	PVC retention policy during helm uninstall when scaled	Retain
naiMonitoring.prometheus.persistentVolumeClaimRetentionPolicy.whenDeleted	PVC retention policy during helm uninstall when deleted	Delete
naiMonitoring.kubeScheduler.serviceEnabled	Enable kube scheduler service	false
naiMonitoring.kubeScheduler.serviceMonitor.enabled	Enable kube scheduler service monitor	false
naiMonitoring.nodeExporter.serviceMonitor.enabled	Enable node exporter service monitor	false
naiMonitoring.dcgmExporter.podLevelMetrics	Enable DCGM exporter pod level metrics	false
naiMonitoring.dcgmExporter.serviceMonitor.enabled	Enable DCGM exporter service monitor	false

Installing or Upgrading Nutanix Enterprise AI on Air-gapped Environments

Install or upgrade Nutanix Enterprise AI on a Kubernetes cluster in an air-gapped environment using Helm charts and the Docker Hub container image library.

Before you begin

> **Note:** If you installed the existing Nutanix Enterprise AI version using the NKP catalog, then upgrade it using the NKP catalog only. For information on how to upgrade Nutanix Enterprise AI using the NKP catalog, see Nutanix Enterprise AI in Workspace in the *Nutanix Kubernetes® Platform Guide*.

- Ensure that your site meets all the requirements mentioned in Nutanix Enterprise AI Requirements on page 11.
- Ensure that you have access to an HTTPS-enabled internal container registry with trusted TLS or SSL certificates.
- Install *Kubectl* and *Helm CLI*.
- Install the following components in your Kubernetes cluster:
  - Envoy Gateway: v1.3.2
  - Cert-manager: v1.16.4 or later version
  - Kserve: 0.15.0 or later version
  - NVIDIA GPU Operator: v25.3.0 or later version on a supported node operating system

    For more information, see NVIDIA documentation.
  - Kube-prometheus-stack: 70.4.2

    > **Note:** For the metrics in Nutanix Enterprise AI to function as expected, ensure that the labels for the service monitors in the nai-system namespace match the respective service labels installed in the Kubernetes cluster.

About this task

To install or upgrade Nutanix Enterprise AI on a Kubernetes cluster in an air-gapped environment, follow these steps:

Procedure

1. From the Nutanix Enterprise AI page, download the following:
   - **NAI Airgap Bundle** containing the container images
   - **NAI Helm Charts** containing the helm charts.

2. Add the *nai-<version>.tar* file containing the container images to a private container registry.

> **Note:** Ensure that the container registry is accessible from the Kubernetes cluster that hosts Nutanix Enterprise AI.

The sample steps to add the container images to such a sample docker container registry are as follows:

a. Load the *.tar* file to your local docker container registry:

```
docker image load -i nai-version.tar
```

Replace *`version`* with the version number of the Nutanix Enterprise AI tar file you downloaded.

> **Note:** Ensure that the container registry and the local directory have the necessary space required to load the container images.

b. (Optional) To view the container images loaded in your local docker container registry, run the following command:

```
$ docker images --format '{{.Repository}}:{{.Tag}}' | grep nai
```

The following is a sample output of the command:

```
nutanix/nai-api:v2.4.0
nutanix/nai-inference-ui:v2.4.0
nutanix/nai-model-processor:v2.4.0
nutanix/nai-iep-operator:v2.4.0
nutanix/nai-tgi:3.3.4-b2485c9
nutanix/nai-kserve-huggingfaceserver:e7295f58
nutanix/nai-kserve-huggingfaceserver:v0.15.2-gpu
nutanix/nai-kserve-huggingfaceserver:v0.15.2
nutanix/nai-kserve-controller:v0.15.0
nutanix/nai-postgres:16.1-alpine
nutanix/nai-kserve-custom-model-server:v2.4.0
```

> **Note:** Ensure that the container images in the output are tagged with the same tag name when you push them to your private registry.

c. Log in to your image registry:

```
docker login image-registry-url
```

Replace *`image-registry-url`* with the URL (without https://) of the image registry where your images are hosted.

d. Tag each image with the version tag and the container repository URL by running the following command for each file separately.

```
docker tag nai-image-name:version image-registry-url/nutanix/nai-image-name:version
```

Replace the following values:

- *nai-image-name* with the name of the container image loaded in your local docker registry.
- *version* with the version of the container image loaded in your local docker registry.
- *image-registry-url* with the URL (without https://) of the image registry where your images are hosted.

e. Push all the tagged image files to the container registry:

```
$ docker push image-registry-url/nutanix/nai-image-name:version
```

Replace the following values:

- *image-registry-url* with the URL (without https://) of the image registry where your images are hosted.
- *nai-image-name* with the name of the container image loaded in your local docker registry.
- *version* with the version of the container image loaded in your local docker registry.

> **Tip:** The following is a sample bash script to tag and push all the files as a batch update to a single image registry. Update the script accordingly based on your setup.
>
> ```
> export IMAGE_REGISTRY_URL=image-registry-url
> for image in $(docker images --format '{{.Repository}}:{{.Tag}}' | grep 'nai'
>  | grep -v $IMAGE_REGISTRY_URL); do
>   docker tag $image $IMAGE_REGISTRY_URL/$(echo $image);
>   docker push $IMAGE_REGISTRY_URL/$(echo $image);
> done
> ```
>
> Replace *image-registry-url* with the URL (without https://) of the image registry where your images are hosted.

3. Extract *nai-core-<version>.tgz* file:

```
$ tar -xvzf nai-core-<version>.tgz
```

The command extracts the following files to the folder:

```
nai-core
### Chart.yaml
### README.md
### aws-values.yaml
### kind-values.yaml
### nke-values.yaml
### nkp-values.yaml
### templates
#   ### _helpers.tpl
#   ### nai-admin
#   ### nai-components
### values.yaml
```

4. If required, push the *nai-core* helm chart to your internal private registry.

5. Based on your setup, make any additional configuration changes to the *nai-core* helm chart.

   For more information, see Nutanix Enterprise AI Configuration Parameters on page 28.

6. Create a *values.yaml* file with required values based on your setup.

   The following is a sample *values.yaml* file.

```
naiIepOperator:
  iepOperatorImage:
    image: image-registry-url/nutanix/nai-iep-operator
    tag: version
  modelProcessorImage:
    image: image-registry-url/nutanix/nai-model-processor
    tag: version
naiInferenceUi:
  naiUiImage:
    image: image-registry-url/nutanix/nai-inference-ui
    tag: version
naiApi:
  naiApiImage:
    image: image-registry-url/nutanix/nai-api
    tag: version
  supportedKserveRuntimeImage: image-registry-url/nutanix/nai-kserve-
huggingfaceserver:kserve-version-gpu
  supportedKserveCPURuntimeImageTag: kserve-version
  supportedKserveGPURuntimeImageTag: kserve-version-gpu
  supportedKserveCustomModelServerRuntimeImage: image-registry-url/nutanix/nai-
kserve-custom-model-server
  supportedKserveCustomModelServerRuntimeImageTag: version
image-registry-url/nutanix/nai-kserve-huggingfaceserver:kserve-version
  supportedTGIImage: image-registry-url/nutanix/nai-tgi
  supportedTGIImageTag: tgi-version
naiDatabase:
  naiDbImage:
    image: image-registry-url/nutanix/nai-postgres:postgres-version
naiMonitoring:
  prometheus:
    image:
      registry: anonymous-access-enabled-image-registry-url
      repository: prometheus-repository-name
      tag: prometheus-version
```

   Replace the following values:

   - `anonymous-access-enabled-image-registry-url` with the URL (without https://) of the anonymous access enabled image registry where your images are hosted.
   - `version` with the version of the container image loaded in your local docker container registry.
   - `kserve-version` with the version of the container image loaded in your local docker container registry.
   - `kserve-version-gpu` with the version of the container image loaded in your local docker container registry.
   - `postgres-version` with the version of the container image loaded in your local docker container registry.
   - `prometheus-repository-name` with the path of the repository that hosts your Prometheus container image.
   - `prometheus-version` with the 2.54.0 version of the Prometheus container image. with the 2.54.0 version of the Prometheus container image.
   - `tgi-version` with the version of the container image loaded in your local docker container registry.

7. Install or upgrade the *nai-core* helm chart helm chart in the *nai-system* namespace:

» If you have the helm chart saved in an internal chart repository:

```
$ helm upgrade --install nai-core nai-core --repo $INTERNAL_REPO_PATH \
-n nai-system --create-namespace \
--set imagePullSecret.credentials.registry=INTERNAL_IMAGE_REGISTRY \
--set imagePullSecret.credentials.username=INTERNAL_IMAGE_REGISTRY_USERNAME \
--set imagePullSecret.credentials.email=INTERNAL_IMAGE_REGISTRY_USERNAME \
--set imagePullSecret.credentials.password=INTERNAL_IMAGE_REGISTRY_PASSWORD \
--set naiApi.storageClassName=NAI_API_RWX_STORAGECLASS \
--set defaultStorageClassName=NAI_DEFAULT_RWO_STORAGECLASS \
-f ./values.yaml \
-f ./ENVIRONMENT-values.yaml --wait
```

» If you have the helm chart saved in your local directory:

```
$ helm upgrade --install nai-core LOCAL_DIRECTORY_PATH \
-n nai-system --create-namespace \
--set imagePullSecret.credentials.registry=INTERNAL_IMAGE_REGISTRY \
--set imagePullSecret.credentials.username=INTERNAL_IMAGE_REGISTRY_USERNAME \
--set imagePullSecret.credentials.email=INTERNAL_IMAGE_REGISTRY_USERNAME \
--set imagePullSecret.credentials.password=INTERNAL_IMAGE_REGISTRY_PASSWORD \
--set naiApi.storageClassName=NAI_API_RWX_STORAGECLASS \
--set defaultStorageClassName=NAI_DEFAULT_RWO_STORAGECLASS \
-f ./LOCAL_DIRECTORY_PATH/values.yaml \
-f ./LOCAL_DIRECTORY_PATH/ENVIRONMENT-values.yaml --wait
```

Replace the following values:

- `INTERNAL_REPO_PATH` with the path of the internal chart repository where you pushed the *nai-core* helm chart.
- `LOCAL_DIRECTORY_PATH` with the path of the local directory where you pushed the *nai-core* helm chart.
- `INTERNAL_IMAGE_REGISTRY` with the URL (without https://) of the image registry where your images are hosted.
- `INTERNAL_IMAGE_REGISTRY_USERNAME` with the username of your image registry where your images are hosted.
- `INTERNAL_IMAGE_REGISTRY_PASSWORD` with the password used to access image registry where your images are hosted.
- `NAI_API_RWX_STORAGECLASS` with the name of the ReadWriteMany (RWX) storage class.

  Note: If you created a dedicated RWX storage class, provide the name as `nai-nfs-storage`. If you did not create a dedicated RWX storage class, provide the name used in `naiApi.storageClassName`.

- `NAI_DEFAULT_RWO_STORAGECLASS` with the name of the ReadWriteOnce (RWO) storage class.
- `values.yaml` template with the file created in Step 7.

- `ENVIRONMENT-values.yaml` with the following templates:
    - NKP clusters: `nkp-values.yaml` template available in the *nai-core* helm chart.
    - Kubernetes clusters on bare metal: `kind-values.yaml` template available in the *nai-core* helm chart.
    - EKS clusters: `aws-values.yaml` template available in the *nai-core* helm chart.
    - AKS clusters: `aks-values.yaml` template created in Step 6 of Setting up an Azure Kubernetes Service Cluster on page 22.
    - GKE clusters: `gke-values.yaml` template created in Step 6 of Setting up a Google Kubernetes Engine Cluster on page 24.

What to do next

- After you upgrade your Nutanix Enterprise AI instance to version 2.3 or later, hibernate and resume any existing CPU-based endpoint to ensure that the endpoint uses the latest runtime with Intel extensions for inferencing. If the CPU generations differ between your production and pre-production environments, your validation results might vary.
- Ensure that all the pods are successfully deployed in the nai-system namespace and are in the ready state by:

  ```
  $ kubectl get pods -n nai-system
  ```

  The following is a sample output of the command:

  ```
  NAME                                            READY   STATUS      RESTARTS   AGE
  nai-api-6b9bc8b9b5-5v62j                         1/1     Running     0
   40h
  nai-api-db-migrate-mxvzl-68blx                   0/1     Completed   0
   39h
  nai-db-0                                         1/1     Running     0
   39h
  nai-iep-model-controller-598d958cd4-k88n6        1/1     Running     0
   40h
  nai-pulse-job-28997285-xhlsb                     0/1     Completed   0
   9h
  nai-ui-7dbd44797b-7gsck                          1/1     Running     0
   40h
  prometheus-nai-0                                 2/2     Running     0
   40h
  ```

Installing or Upgrading Nutanix Enterprise AI with a Self-managed PostgreSQL Database

Install Nutanix Enterprise AI on a Kubernetes cluster with a self-hosted or managed PostgreSQL database server.

Before you begin

- Configure a database that runs PostgreSQL version 16.1 or 16.8.
- Ensure that POSTGRES_HOST_URL has a fully qualified domain name (FQDN) that can be resolved by the DNS service in the Kubernetes cluster hosting Nutanix Enterprise AI.
- Ensure that your site meets all the requirements mentioned in Nutanix Enterprise AI Requirements on page 11.
- Ensure that you set up your cluster based on the requirements mentioned in Kubernetes Cluster Setup on page 19.

- Log in to the Nutanix Support portal and generate a Docker Hub access token. For more information, see Generating Nutanix Docker Hub Access Tokens.

  The access token is required to obtain the Helm chart and the Nutanix Enterprise AI container artifacts from the Nutanix Enterprise AI page.
- Install *Kubectl* and *Helm CLI*.
- To make any additional configuration changes to the *nai-core* helm chart, see Nutanix Enterprise AI Configuration Parameters on page 28.

About this task

> **Important:** Installing or upgrading Nutanix Enterprise AI with a self-managed PostgreSQL database is an Experimental feature. Do not use Experimental features in a production environment. To participate in the Experimental program, contact your Nutanix product team. Your use of any Experimental technology is subject to the Nutanix License and Service Agreement.

If your environment includes a PostgreSQL database, either self-hosted or a managed service like Amazon RDS, you can deploy Nutanix Enterprise AI to work with the database. This eliminates the need to manage additional database servers, reduces the complexity of security policies and firewall rules, enforces stricter quality checks on enterprise-specific images to ensure reliability and security, and enables instance replication for robust disaster recovery.

To install or upgrade Nutanix Enterprise AI on a Kubernetes cluster with a PostgreSQL database, follow these steps:

Procedure

1. Add and update the Nutanix helm repository, which contains the *nai-core* helm chart:

   ```
   $ helm repo add ntnx-charts https://nutanix.github.io/helm-releases && helm repo
    update ntnx-charts
   ```

2. Search for the version of the *nai-core* helm chart available for installation in the Nutanix helm repository:

   ```
   $ helm search repo ntnx-charts/nai-core --versions
   ```

3. Pull the *nai-core* helm chart and extract it to your local repository:

   ```
   $ helm pull ntnx-charts/nai-core --version=nai-core-version --untar=true
   ```

   Replace `nai-core-version` with the version of *nai-core* helm chart that you obtained in Step 2.

4. Install or upgrade the *nai-core* helm chart in the *nai-system* namespace:

   ```
   $ helm upgrade --install nai-core ntnx-charts/nai-core --version=NAI_CORE_VERSION -n
    nai-system --create-namespace --wait \
           --set imagePullSecret.credentials.username=DOCKER_USERNAME \
           --set imagePullSecret.credentials.email=DOCKER_USERNAME \
           --set imagePullSecret.credentials.password=DOCKER_API_KEY \
           --insecure-skip-tls-verify \
           --set naiApi.storageClassName=NAI_API_RWX_STORAGECLASS \
           --set defaultStorageClassName=NAI_DEFAULT_RWO_STORAGECLASS \
           --set naiDatabase.external=true \
           --set naiDatabase.postgresql.postgresDatabase=CONFIGURED_DB_NAME \
           --set naiDatabase.postgresql.postgresUser=POSTGRES_USERNAME \
           --set naiDatabase.postgresql.postgresPassword=POSTGRES_PASSWORD \
           --set naiDatabase.postgresql.postgresHost=POSTGRES_HOST_URL \
   ```

```
        -f ./nai-core/values.yaml
```

Replace the following values:

- `NAI_CORE_VERSION` with the version of *nai-core* helm chart that you obtained in Step 2.
- `DOCKER_USERNAME` with the username of your Docker account.
- `DOCKER_API_KEY` with the API key or the access token used to access your Docker account.
- `NAI_API_RWX_STORAGECLASS` with the name of the ReadWriteMany (RWX) storage class.

  > **Note:** If you created a dedicated RWX storage class, provide the name as `nai-nfs-storage`. If you did not create a dedicated RWX storage class, provide the name used in `naiApi.storageClassName`.

- `NAI_DEFAULT_RWO_STORAGECLASS` with the name of the ReadWriteOnce (RWO) storage class.
- `CONFIGURED_DB_NAME` with the name of the PostgreSQL database.
- `POSTGRES_USERNAME` with the username of your PostgreSQL database.
- `POSTGRES_PASSWORD` with the password used to access your PostgreSQL database.
- `POSTGRES_HOST_URL` with the URL of the PostgreSQL database.
- `values.yaml` with the following templates:
  - EKS clusters: `aws-values.yaml` template available in the *nai-core* helm chart.
  - NKP clusters: `nkp-values.yaml` template available in the *nai-core* helm chart.
  - AKS clusters: `aks-values.yaml` template created in Step 6 of *Setting up an Azure Kubernetes Service Cluster*.
  - GKE clusters: `gke-values.yaml` template created in Step 6 of *Setting up a Google Kubernetes Engine Cluster*.

  > **Note:** The `sslmode` parameter is not supported.

What to do next

- After you upgrade your Nutanix Enterprise AI instance to version 2.3 or later, hibernate and resume any existing CPU-based endpoint to ensure that the endpoint uses the latest runtime with Intel extensions for inferencing. If the CPU generations differ between your production and pre-production environments, your validation results might vary.
- Ensure that all the pods are successfully deployed in the nai-system namespace and are in the ready state by:

  ```
  $ kubectl get pods -n nai-system
  ```

  The following is a sample output of the command:

  ```
  NAME                                            READY   STATUS      RESTARTS   AGE
  nai-api-6c85c95866-bkwm2                          1/1     Running     0
   23m
  nai-api-db-migrate-h5sqd-xxhsr                    0/1     Completed   0
   23m
  nai-iep-model-controller-85df96896d-2zzrf         1/1     Running     0
   23m
  nai-ui-77f6b4ddc6-nhk4c                           1/1     Running     0
   23m
  prometheus-nai-0                                  2/2     Running     0
   23m
  ```

Logging in to Nutanix Enterprise AI

Log in to Nutanix Enterprise AI.

About this task

To log in to Nutanix Enterprise AI, follow these steps:

Procedure

1. Open a web browser, enter the configured fully qualified domain name (FQDN) or the external IP address of the Envoy Ingress Gateway service in the address field, and press Enter.
   The Nutanix Enterprise AI login page opens.
2. Enter your login credentials and click **Login**.

   Upon successful login, the system displays the Nutanix Enterprise AI dashboard.

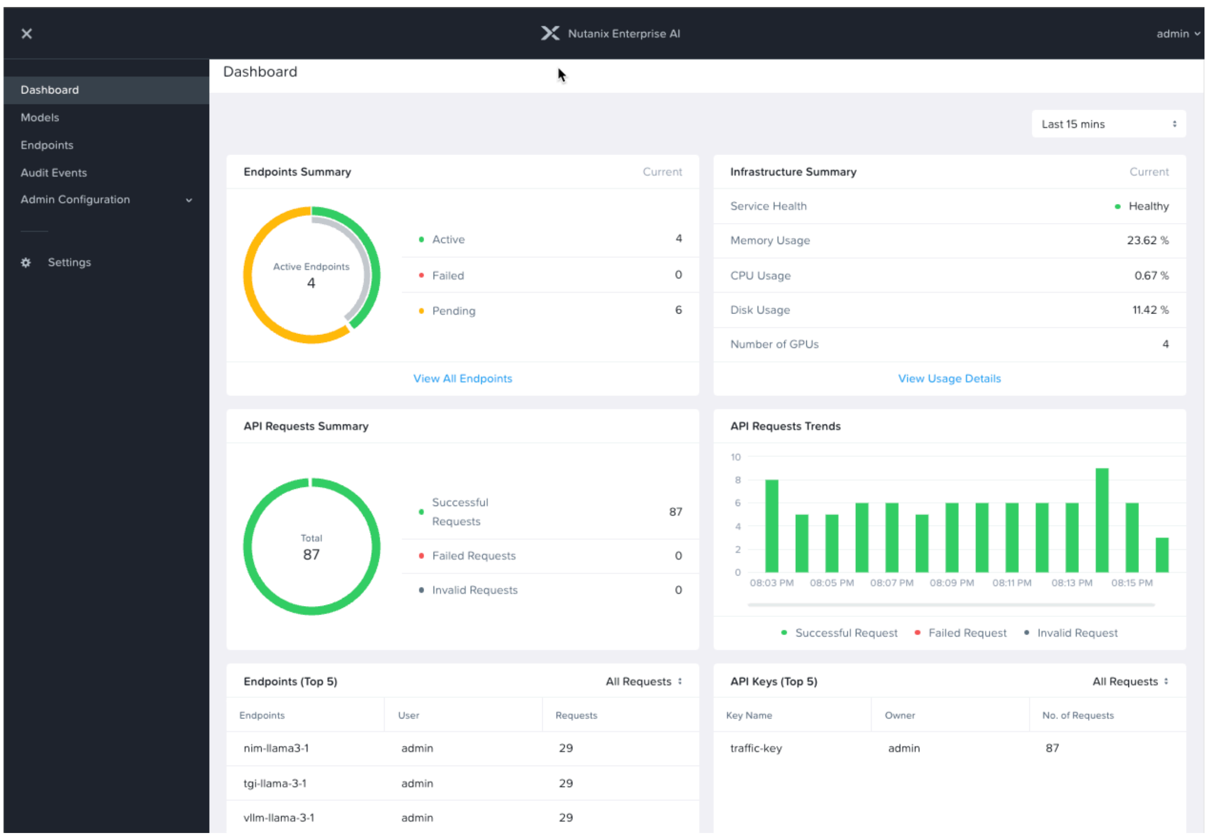

Figure 3: Dashboard inNutanix Enterprise AI

Important:

- When you log in to Nutanix Enterprise AI for the first time, the system prompts you to change the default password. Enter a new password in the **Type password** and **Retype password** fields, then click **Submit**.

  After you successfully change the password, the system synchronizes the new password across the database.
- If you are logging in as a Super Admin user (user name admin and default password Nutanix.123) for the first time and if an end-user license agreement (EULA) screen appears (typically on the first

login or if the end-user license agreement has changed since the last login) indicating that you have not acknowledged the current EULA, do the following:

1. Read the license agreement (on the left).
2. Enter the appropriate information in the Name, Company, and Job Title fields (on the right).
3. Select the **I have read and agree to the terms and conditions** checkbox.
4. Click **Accept**.

Changing the Language Settings in Nutanix Enterprise AI

Change the language settings in Nutanix Enterprise AI. Nutanix Enterprise AI supports English and Japanese.

About this task

A user with a Super Admin role can change the user interface language from English (the default) to Japanese, or vice versa.

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Settings**.
   The **Third Party Credentials** page opens.
3. Click the **Language Settings** tab.
   The system displays the current language settings.
4. Click **Edit**.
   The system displays the **Edit Language Settings** dialog box.
5. From the **Language** dropdown menu, select the language.
   The system applies the selected language to the Nutanix Enterprise AI user interface for all other users in the instance.
6. Instruct the users to reload their Nutanix Enterprise AI instance.
   Users can view the changes.

VIEWING THE DASHBOARD IN NUTANIX ENTERPRISE AI

The dashboard provides a dynamic summary view of all the endpoints, the API requests, and the API keys.

About this task

To access the **Dashboard** page, follow these steps.

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Dashboard**.
   The system displays the dashboard with the widgets. For information on the widgets that appear on the dashboard, see Dashboard Widgets on page 42.

Dashboard Widgets

The following tables describe the widgets that appear on the Nutanix Enterprise AI dashboard.

Table 16: Dashboard Widgets

Widget Name	Description
Endpoints Summary	Displays the total number of endpoints based on their status, such as **Active**, **Hibernated**, **Failed**, and **Pending**. Click **View All Endpoints** to go to the **Endpoints** page. For more information, see Viewing Endpoints in Nutanix Enterprise AI on page 66.
Infrastructure Summary	Displays the summary of the health and infrastructure usage of the Kubernetes cluster that hosts Nutanix Enterprise AI. This widget displays the health status of all the active inference endpoints, the memory usage, CPU usage, disk usage, and the number of GPUs in the cluster. **Important:** The number of GPUs displayed might be incorrect if the GPU operator pods in your Kubernetes cluster are unhealthy. To fix these unhealthy pods, contact NVIDIA support. Click **View Details** to view the usage statistics of the cluster and the individual nodes in detail. For more information, see Cluster Usage Metrics on page 91 and Node Usage Metrics on page 92.
Requests Summary	Displays the total number of API requests and API requests based on status, such as **Successful Requests**,**Failed Requests**, and **Invalid Requests**.

Widget Name	Description
Requests Trends	Displays the trend in API requests over a period of time, based on status, such as **Successful Requests**,**Failed Requests**, and **Invalid Requests**.
Endpoints (Top 5)	Displays the five most used inference endpoints.
API Keys (Top 5)	Displays the five most used API keys based on the number of requests or tokens used such as **All Requests**, **Successful Requests**,**Failed Requests**, **Input Token Usage**, **Output Token Usage**.

ADDING A HUGGING FACE TOKEN

To download a large language model (LLM) from Hugging Face to Nutanix Enterprise AI, you must add a Hugging Face access token to Nutanix Enterprise AI.

Before you begin

- Create a Hugging Face account. For more information, see Hugging Face documentation.
- Create an access token in your Hugging Face account. For more information, see *Generating a User Access Token* in Hugging Face documentation.

  > **Note:** If you create a fine-grained access token in Hugging Face, provide read access permission to the repository of the model hosted in Hugging Face to ensure that the model can be imported to Nutanix Enterprise AI. For more information, see *User access tokens best practises* in Hugging Face documentation.

About this task

To add a new Hugging Face token to Nutanix Enterprise AI, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Settings**.

   The **Third Party Credentials** page opens.
3. Click the **Add** option associated with **Hugging Face Model Hub Token**.

   The **Add Hugging Face Account token** dialog box opens.
4. In the **Token Name** field, enter the Hugging Face token name.
5. In the **Access Token** field, enter the token key obtained from Hugging Face.
6. Click **Add**.
   The Hugging Face token is added to Nutanix Enterprise AI.

Replacing a Hugging Face Token

You can add a new Hugging Face token to replace the existing token in Nutanix Enterprise AI.

Before you begin

- Create a Hugging Face account. For more information, see Hugging Face documentation.
- Create an access token in your Hugging Face account. For more information, see *Generating a User Access Token* in Hugging Face documentation.

  > **Note:** If you create a fine-grained access token in Hugging Face, provide read access permission to the repository of the model hosted in Hugging Face to ensure that the model can be imported to Nutanix Enterprise AI. For more information, see *User access tokens best practices* in Hugging Face documentation.

About this task

To replace the existing token in Nutanix Enterprise AI, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation bar, click **Settings**.

   The **Third Party Credentials** page opens.
3. From the **Hugging Face Model Hub Token** section, click **Update**.

   The **Manage Hugging Face Model Hub Token** dialog box opens.
4. In the **Token Name** field, enter the new token name.
5. In the **Access Token** field, enter the new token obtained from Hugging Face.
6. Click **Update**.
   The new Hugging Face token is added to Nutanix Enterprise AI.

ADDING AN NVIDIA NGC PERSONAL KEY

To download NVIDIA NIMs from NVIDIA NGC Catalog to Nutanix Enterprise AI, you must add an NVIDIA NGC Personal Key to Nutanix Enterprise AI.

Before you begin

- Ensure that you have an NGC account with an active subscription. For more information, see NVIDIA Documentation Hub.
- Generate an NVIDIA NGC Personal Key. For more information, see the *Personal API Key* section in the *NGC User Guide* listed in NVIDIA Documentation Hub.

> **Note:** After you generate an NVIDIA NGC Personal Key, add *NGC Catalog* services to the personal key to ensure that the NIM can be imported to Nutanix Enterprise AI. For more information, see the *Personal API Key* section in the *NGC User Guide* listed in NVIDIA Documentation Hub.

About this task

To add a new NGC Personal Key to Nutanix Enterprise AI, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Settings**.

   The **Third Party Credentials** page opens.
3. Click the **Add** option associated with **NVIDIA NGC Personal Key**.

   The **Manage NVIDIA NGC Personal Key** dialog box opens.
4. In the **Key Name** field, enter the personal key name.
5. In the **Key Value** field, enter the personal key obtained from NGC.
6. Click **Add**.

   The personal key is added to Nutanix Enterprise AI.

Replacing an NVIDIA NGC Personal Key

You can add a new NVIDIA NGC Personal Key to replace the existing key in Nutanix Enterprise AI.

Before you begin

- Ensure that you have an NGC account with an active subscription. For more information, see NVIDIA Documentation Hub.
- Generate an NVIDIA NGC Personal Key. For more information, see the *Personal API Key* section in the *NGC User Guide* listed in NVIDIA Documentation Hub.

> **Note:** After you generate an NVIDIA NGC Personal Key, add *NGC Catalog* services to the personal key to ensure that the NIM can be imported to Nutanix Enterprise AI. For more information, see the *Personal API Key* section in the *NGC User Guide* listed in NVIDIA Documentation Hub.

About this task

To replace the existing NVIDIA NGC Personal Key in Nutanix Enterprise AI, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation bar, click **Settings**.

   The **Third Party Credentials** page opens.
3. From the **NVIDIA NGC Personal Key** section, click **Update**.

   The **Manage NVIDIA NGC Personal Key** dialog box opens.
4. In the **Key Name** field, enter the new personal key name.
5. In the **Key Value** field, enter the new personal key obtained from NGC.
6. Click **Update**.

   The new key is added to Nutanix Enterprise AI.

LARGE LANGUAGE MODELS IN NUTANIX ENTERPRISE AI

You can select and import text-based generative LLMs from Hugging Face or NVIDIA NGC Catalog to Nutanix Enterprise AI. These LLMs are displayed in the **Models** page.

Nutanix Enterprise AI supports importing an LLM in the following ways:

Import a pre-validated LLM from Hugging Face

: You can directly import a pre-validated LLM from Hugging Face. For more information, see Importing a Large Language Model from Hugging Face on page 56. For information on the LLMs that can be imported directly, see Pre-validated Models on page 51.

Import a pre-validated NIM from NVIDIA NGC Catalog

: You can directly import a NIM from NVIDIA NGC Catalog. For more information, see Importing NVIDIA NIMs from NVIDIA NGC Catalog on page 62. For information on the pre-validated NIMs that can be imported directly, see Pre-validated Models on page 51.

Import an LLM from Hugging Face using Model URL

: You can import an LLM not validated by Nutanix directly from Hugging Face by adding the URL of the model in Nutanix Enterprise AI. For more information, see Importing a Large Language Model from Hugging Face using Model URL or ID on page 58.

Import a pre-validated LLM from Hugging Face manually

: If Nutanix Enterprise AI is deployed in a dark site, you can download a pre-validated LLM from Hugging Face to your local storage, Network File System (NFS) file share, or a S3 compatible Object bucket and then import it to Nutanix Enterprise AI. Dark sites are primarily on-premises installations that do not have access to the internet.

: For information on how to manually import an LLM, see Importing a Large Language Model Manually from Hugging Face on page 60.

Import a custom LLM

: A user who has advanced knowledge about LLM operations can import a custom LLM from local storage, Network File System (NFS) file share, or an S3 compatible Object bucket to Nutanix Enterprise AI. A customized LLM might be a fine-tuned version or the latest version of an existing LLM and might resemble the pre-validated LLMs in its architecture. However, Nutanix does not validate a custom LLM when you import it to Nutanix Enterprise AI.

: For information on how to import a custom LLM, see Importing a Large Language Model Manually from Hugging Face on page 60.

Viewing Imported Large Language Models

The **Models** page displays all the large language models (LLMs) that are imported to Nutanix Enterprise AI.

About this task

To view your imported LLMs, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.

2. From the left navigation pane, select **Models**.
   The **Models** page opens displaying a summary of all the LLMs imported to Nutanix Enterprise AI. For information on the LLM attributes displayed in the page, see Imported LLMs Attributes on page 49.

Imported LLMs Attributes

The following table describes the LLM attributes that appear on the **Models** page.

Table 17: Imported LLMs Attributes - Description

<table>
<tr><th>Attributes</th><th>Description</th><th>Values</th></tr>
<tr><td>Model Instance Name</td><td>Displays the

model name that you provide while importing the model.
Link to the Model Details page. The Model Details page displays the following:

Model Instance Name: Displays the model name that you provide while importing the model.
Model: Displays thename of the model
Developer: developer of the model
Repo version: Displays the version of the model
Import Mode: Displays the mode and source from where the model was imported.

If you manually uploaded the model, hovering over the hover info icon displays the Storage Provider, Server IP, NFS Export Path, and the Directory Path for the Model.
Type: Displays the type of the model
Model Size: Displays the size of the model
Imported By: Displays the user who imported the model
Imported On: Displays the date when the model was imported.
Status: Displays the status.

</td><td>LLM instance name and the link to the Model Details page</td></tr>
</table>

Attributes	Description	Values
Model	Displays the following: • name of the model as per the Hugging Face or NVIDIA database. • live links of Hugging Face and NIM models that point to their source pages on Hugging Face Hub or NVIDIA NGC.	LLM name or link to the source page
Developer	Displays the name of the company that developed the model.	Name of developer
Import Mode	Displays the method used to import the model.	Hugging Face, NVIDIA NIM, Manual Upload
Type	Displays the type of imported model.	Embedding, Reranker, Safety, Text Generation
Imported By	Displays the name of the user who imported the model.	User name
Status	Displays the current status of the model.	Ready, Failed, Pending, Processing

Pre-validated Models

This section describes the models in Hugging Face and NVIDIA NIM format, which Nutanix tested and validated to run successfully on Nutanix Enterprise AI. The CPU and memory requirements for these models are automatically populated when deployed as an endpoint.

The following table lists the pre-validated models and the size of each model.

> **Note:** In addition to the pre-validated models listed in the following table, Nutanix Enterprise AI also supports importing unvalidated NVIDIA NIMs or custom LLM models. The architecture of a custom model might resemble the architecture of a listed pre-validated model. However, Nutanix does not validate these models when you import them to Nutanix Enterprise AI.

Table 18: Pre-validated Models

Model Hub	Provider	Model	Model Type	Model Size (GiB)
Hugging Face	AI21 Labs	ai21labs/AI21-Jamba-1.5-Mini	Text Generation	110
	Google	google/gemma-2-2b-it	Text Generation	10
		google/gemma-2-9b-it	Text Generation	20
		google/vit-base-patch16-224	Image Classification	4

Model Hub	Provider	Model	Model Type	Model Size (GiB)
	Meta	meta-llama/Llama-2-13b-chat-hf	Text Generation	60
		meta-llama/Llama-3.2-3b-Instruct	Text Generation	20
		meta-llama/Llama-3.2-1B-Instruct	Text Generation	10
		meta-llama/Llama-3.3-70B-Instruct	Text Generation	290
		meta-llama/Meta-Llama-3.1-8B-Instruct	Text Generation	40
		meta-llama/Meta-Llama-3.1-70B-Instruct	Text Generation	290
		meta-llama/CodeLlama-7b-Instruct-hf	Text Generation	30
		meta-llama/CodeLlama-13b-Instruct-hf	Text Generation	60
		meta-llama/CodeLlama-34b-Instruct-hf	Text Generation	140
		meta-llama/CodeLlama-70b-Instruct-hf	Text Generation	280
		meta-llama/Llama-3.2-11B-Vision-Instruct	Vision	55
		meta-llama/Llama-3.2-90B-Vision-Instruct	Vision	320
		meta-llama/Llama-4-Scout-17B-16E-Instruct	Text Generation	250
		meta-llama/Llama-Guard-3-8B	Safety	17
	Mistral AI	mistralai/Mistral-7B-Instruct-v0.3	Text Generation	30

Model Hub	Provider	Model	Model Type	Model Size (GiB)
		mistralai/Mixtral-8x7B-Instruct-v0.1	Text Generation	200
		mistralai/Mixtral-8x22B-Instruct-v0.1	Text Generation	290
		mistralai/Mistral-Nemo-Instruct-2407	Text Generation	50
	IBM	ibm-granite/granite-embedding-107m-multilingual	Embedding	2
	Cross-Encoder	cross-encoder/ms-marco-MiniLM-L6-v2	Reranker	4
	Facebook	facebook/deit-base-distilled-patch16-224	Image Classification	4
	Stability AI	stable-diffusion-v1-5/stable-diffusion-v1-5	Image Generation	40
	Unsloth	unsloth/Llama-3.3-70B-Instruct-bnb-4bit	Text Generation	50
NVIDIA NGC Catalog	NVIDIA	llama-3.1-8b-instruct	Text Generation	50
		llama-3.1-70b-instruct	Text Generation	160
		llama-3.1-nemoguard-8b-content-safety	Safety	50
		llama-3.2-nv-embedqa-1b-v2	Embedding	5
		llama-3.2-nv-rerankqa-1b-v2	Reranker	5
		llama-3.3-70b-instruct	Text Generation	160
		llama-3.3-nemotron-super-49b-v1	Text Generation	120
		llama-3.1-swallow-8b-instruct-v0	Text Generation	50
		llama-3.1-nemoguard-8b-topic-control	Safety	50

Model Hub	Provider	Model	Model Type	Model Size (GiB)
		mixtral-8x7b-instruct-v01	Text Generation	110
		mistral-7b-instruct-v0	Text Generation	50
		phi-3-mini-4k-instruct	Text Generation	10
		black-forest-labs/flux.1-dev **Note:** To use the model, you must have a valid Hugging Face token added to Nutanix Enterprise AI, and that token must have access permissions for the model on Hugging Face.	Image Generation	40
		Mistral-nemo-12b-instruct	Text Generation	80
		Llama-3.2-90b-vision-instruct	Vision	200
		Llama-3.1-70b-instruct-pb24h2	Text Generation	160
		Llama-3.1-swallow-8b-instruct-v0.1	Text Generation	50
		Llama-3.1-nemotron-70b-instruct	Text Generation	160
		Llama-3.1-8b-instruct-pb24h2	Text Generation	50
		Mistral-7b-instruct-v0.3	Text Generation	50
		Mixtral-8x7B-Instruct-v0.1	Text Generation	110
		gpt-oss-20b	Text Generation	60
		gpt-oss-120b	Text Generation	210

Configuring Access to Models

Configure AI/ML User (user) access to models.

Before you begin

Add users. For more information, see Creating Users in Nutanix Enterprise AI on page 87.

About this task

The default configuration to view and access models is as follows:

- If you upgraded Nutanix Enterprise AI, users can view all the models from prior versions. New models are disabled.
- If you installed Nutanix Enterprise AI for the first time, all the models are disabled by default.

AI/ML Admins (admin) can either use the default configuration or configure permissions to view and download models.

To restrict AI/ML User (user) access to models in the catalog, complete the following steps:

Procedure

1. Log in to Nutanix Enterprise AI as an admin.
2. From the left navigation pane, select **Models** > **Model Access Control**.

   The **Model Access Control** page opens.

   > **Note:** The default configuration of **Allow direct download using model URL** and **Allow Manual Upload** are as follows:
   >
   > - Both are enabled if:
   >   - You upgraded Nutanix Enterprise AI
   >   - You have uploaded models.
   > - Both are disabled if :
   >   - You installed Nutanix Enterprise AI
   >   - You have not uploaded models.

3. (Optional) To restrict the models that users can download:

    a. In **Access Control for Catalogs**, select **Modify Allowed List** for the required catalog.

    b. (Optional) To allow all the validated models, select **Allow all validated models**.

    c. (Optional) To select only specific models, do the following:

        1. Select **Allow Specific models**.
        2. Select the required models.

    d. Click **Save**.

        - users cannot import the restricted models.
        - users can continue to use the models they downloaded before you applied this restriction. You will see a notification to delete the restricted models in the **Models List** screen and the **Endpoints List** screen.
        - users cannot create new endpoints on previously imported and now disabled models.

    e. To meet regulations, inform users to manually delete the restricted models and endpoints.

4. (Optional) To allow users to download Hugging Face Model hub models that are not in the catalog enable **Import Model using model URL** .

5. (Optional) To prevent users from downloading models Hugging Face Model hub that are not in the catalog disable **Import Model using model URL** .

6. (Optional) To allow users to upload a model from a file share or bucket, enable **Allow Manual Upload**.

7. (Optional) To allow only admins to upload a model,

    a. Disable **Allow Manual Upload**.

    b. Click **Disable**.

    After you disable, only admins can upload models or catalog models. Users cannot upload models or catalog models.

    Admin can import disabled models. However, a warning that `Some models are not compliant with the current organization policy.` is displayed in the **Models List** page.

What to do next

- Inform users that they cannot download the restricted models in future.
- Inform users to stop using the restricted models that they downloaded before you applied this restriction. To stop using restricted models,

    1. Delete the endpoints. For more information, see Deleting an Endpoint on page 74.
    2. Delete the restricted models. For more information, see Deleting a Large Language Model on page 63.

Importing a Large Language Model from Hugging Face

Import a pre-validated LLM from Hugging Face to Nutanix Enterprise AI.

Before you begin

- Ensure that you accept the usability terms and licenses of the LLM and agree to share your contact information (email address and username) with the repository authors of the model in Hugging Face to access the repository and import the LLM to Nutanix Enterprise AI.

- Create an access token in Hugging Face and add it to Nutanix Enterprise AI. For more information, see Adding a Hugging Face Token on page 44.

> **Note:** If you create a fine-grained access token in Hugging Face, provide read access permission to the repository of the model hosted in Hugging Face to ensure that the model can be imported to Nutanix Enterprise AI. For more information, see *User access tokens best practices* in Hugging Face documentation.

About this task

To import an LLM from Hugging Face, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Models**.
   The system displays the **Models** page.
3. Click **Import Models** > **From Hugging Face Model Hub**.

   > **Note:** If you are logging in for the first time, from the **Models** page, click **Import Model** > **View Validated Models**.

   The **Import Model - Hugging Face Model Hub** page opens, displaying all the LLMs validated to run on Nutanix Enterprise AI.
4. (Optional) In the **Search by Model Name** field, type the LLM name to search for an LLM.
   The system lists the matching model names as you type.
5. (Optional) To filter and view the LLMs based on the model type, select the model type from the **All Models** dropdown menu.
6. Select an LLM and click **Import**.

   > **Note:**
   >
   > - To go to an LLM's Hugging Face repository, click the name of the LLM.
   > - If you did not add the Hugging Face access token to Nutanix Enterprise AI, the system prompts you to do so. Add the access token to import the LLM. For more information, see Adding a Hugging Face Token on page 44.

   The **Import Model** dialog box opens.
7. In the **Model Instance Name** field, enter a name for the LLM.

   Nutanix recommends that you use the actual name of the LLM, suffixed with an identifier that is meaningful to you.
8. Click **Import**.
   The imported LLM is displayed on the **Models** page.

What to do next

After you initiate an import, you can view the status of the import on the **Models** page. The system displays one of the following states for the import operation:

- **Ready**: The LLM is imported and ready to use.

> **Note:** You can create an endpoint from an LLM only if the status displays **Ready**.

- **Failed**: The import failed due to insufficient storage, unauthorized Hugging Face token, or LLM repository read access restriction for the access token.
- **Pending**: The system is waiting for the resources required to save the LLM.

  For example, if the required storage space is not available, Nutanix Enterprise AI maintains a **Pending** status until the storage space becomes available.
- **Processing**: The system is downloading the LLM from Hugging Face.

After you successfully import an LLM to Nutanix Enterprise AI, you can deploy the LLM to an AI endpoint. Complete the following steps:

1. Select the model.
2. Click **Actions** > **Create Endpoint** .

   The **Create Endpoint** action is enabled only for active models and the models you import. You cannot deploy models created by other users.
3. The **Create Endpoint** screen is displayed and the **Model Instance Name** field displays the imported model.
4. Create the endpoint.

   For more information, see Creating an Endpoint on page 70.

Importing a Large Language Model from Hugging Face using Model URL or ID

Import an LLM not validated by Nutanix directly from Hugging Face, by adding the URL of the model in Nutanix Enterprise AI.

Before you begin

- Create an access token in Hugging Face and add it to Nutanix Enterprise AI. For more information, see Adding a Hugging Face Token on page 44.

> **Note:** If you create a fine-grained access token in Hugging Face, provide read access permission to the repository of the model hosted in Hugging Face to ensure that the model can be imported to Nutanix Enterprise AI. For more information, see *User access tokens best practices* in Hugging Face documentation.

- Ensure that you accept the usability terms and licenses of the LLM and agree to share your contact information (email address and username) with the repository authors of the model in Hugging Face to access the repository and import the LLM to Nutanix Enterprise AI.

About this task

Unlike the manual import method, this method does not require you to manually provision storage, such as a Network File System (NFS) file share or an S3-compatible Object bucket. This import method avoids downloading and storing the LLM locally, making it efficient for environments with limited storage space. However, Nutanix does not test and validate an LLM when you import it using the model URL.

To import an LLM from Hugging Face using model URL, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.

2. From the left navigation pane, select **Models**.
   The system displays the **Models** page.

3. Click **Import Models** > **From Hugging Face Model Hub**.
   The **Import Model - Hugging Face Model Hub** page opens, displaying all the LLMs validated to run on Nutanix Enterprise AI.

4. Click **Import using Model URL**.
   The system displays the **Import Model using Hugging Face Model URL** dialog box.

5. In the **Model URL** field, enter a valid Hugging Face URL or ID of the LLM.

   You can specify the prefix for the URL,as huggingface.com/, hf.co/, or huggingface.co/.

6. In the **Model Instance Name** field, enter a name for the LLM.

   Nutanix recommends that you use the actual name of the LLM, suffixed with an identifier that is meaningful to you.

7. (Optional) In the **Developer (Optional)** field, enter the name of the LLM developer.

8. From the **Model Type** dropdown menu, select the LLM type.

9. Click **Import**.
   The system prompts you to confirm the import action.

10. In the field provided, type **confirm** and click **Import**.
    The system displays the imported LLM on the **Models** page.

What to do next

After you initiate an import, you can view the status of the import on the **Models** page. The system displays one of the following states for the import operation:

- **Ready**: The LLM is imported and ready to use.

  > **Note:** You can create an endpoint from an LLM only if the status displays **Ready**.

- **Failed**: The import failed due to insufficient storage, unauthorized Hugging Face token, or LLM repository read access restriction for the access token.

- **Pending**: The system is waiting for the resources required to save the LLM.

  For example, if the required storage space is not available, Nutanix Enterprise AI maintains a **Pending** status until the storage space becomes available.

- **Processing**: The system is downloading the LLM from Hugging Face.

After you successfully import an LLM to Nutanix Enterprise AI, you can deploy the LLM to an AI endpoint. Complete the following steps:

1. Select the model.
2. Click **Actions** > **Create Endpoint** .

   The **Create Endpoint** action is enabled only for active models and the models you import. You cannot deploy models created by other users.
3. The **Create Endpoint** screen is displayed and the **Model Instance Name** field displays the imported model.
4. Create the endpoint.

   For more information, see Creating an Endpoint on page 70.

Importing a Large Language Model Manually from Hugging Face

Manually import a pre-validated LLM from Hugging Face or a custom LLM to Nutanix Enterprise AI.

Before you begin

Ensure that you download the pre-validated LLM from Hugging Face in the original file format and save it to your local storage, Network File System (NFS) file share, or an S3 compatible Object bucket. For information on how to download an LLM from Hugging Face, see *Downloading models* in Hugging Face documentation.

About this task

To manually import an LLM from your local storage, NFS file share, or a S3 compatible Object bucket, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Models**.
   The system displays the **Models** page.
3. Click **Import Models** > **Using Manual Import**.

   > **Note:** If you are logging in for the first time, click **Import Model** > **Import Manually** from the **Models** page.

   The **Manual Upload** page opens.
4. From the **Choose a Model Type** section, choose one of the following:

   » **Pre-validated Model**: Select this option to upload a pre-validated LLM in the original format as downloaded from Hugging Face.

   » **Custom Model**: Select this option to upload a custom LLM.

   > **Note:** Nutanix does not validate a custom LLM when you import it to Nutanix Enterprise AI.
5. From the **Model** dropdown menu, select the pre-validated LLM.

   > **Note:** This field appears only if you select **Pre-validated Model** in Step 4.
6. In the **Model Instance Name** field, enter a name for the LLM.

   Nutanix recommends that you use the actual name of the LLM, suffixed with an identifier that is meaningful to you.
7. From the **Model Type** dropdown menu, select the LLM type.
8. In the **Model Size** field, enter the storage size required to store the downloaded files.

   > **Note:** This field appears only if you select **Custom Model** in Step 4.
9. (Optional) In the **Developer (Optional)** field, enter the name of the LLM developer.

   > **Note:** This field appears only if you select **Custom Model** in Step 4.

10. From the **Location** dropdown menu, select the appropriate option:

    » **File Share**: If the LLM is saved in an NFS file share.

    » **Bucket**: If the LLM is saved in a S3 compatible Object bucket.

11. If you select **File Share** in Step 10, enter the following details:

    - **File Server Address**: Enter the fully qualified domain name (FQDN) or the IP address of the Nutanix Files server.
    - **NFS Export Path**: Enter the share path to the NFS export.
    - **Directory Path for the Model**: Enter the directory path to the NFS export where you have saved the LLM.

      > **Note:** Nutanix does not validate the NFS configuration in your cluster. An incorrect NFS configuration results in an LLM import failure.

12. If you select **Bucket** in Step 10, enter the following details:

    - **Service Host**: Enter the complete URL or the IP address of the endpoint used to tier the objects.

      For example, `example.buckets.company.com`.
    - **Bucket Name**: Enter the name of the S3-compatible Object bucket within the service host to which the objects must tier out.
    - **Model Path**: Enter the prefix of the S3-compatible Object bucket key name.
    - **Access Key**: Enter the access key of the S3-compatible Object bucket owner.
    - **Secret Key**: Enter the secret key of the S3-compatible Object bucket owner.

13. Click **Upload**.
    The system displays the imported LLM on the **Models** page.

What to do next

After you initiate an import, you can view the status of the import on the **Models** page. The system displays one of the following states for the import operation:

- **Ready**: The LLM is imported and ready to use.

  > **Note:** You can create an endpoint from an LLM only if the status displays **Ready**.

- **Failed**: The import failed due to insufficient storage, unauthorized Hugging Face token, or LLM repository read access restriction for the access token.
- **Pending**: The system is waiting for the resources required to save the LLM.

  For example, if the required storage space is not available, Nutanix Enterprise AI maintains a **Pending** status until the storage space becomes available.
- **Processing**: The system is downloading the LLM from Hugging Face.

After you successfully import an LLM to Nutanix Enterprise AI, you can deploy the LLM to an AI endpoint. Complete the following steps:

1. Select the model.

2. Click **Actions** > **Create Endpoint** .

   The **Create Endpoint** action is enabled only for ready models and the models you import. You cannot deploy models created by other users.
3. The **Create Endpoint** screen is displayed and the **Model Instance Name** field displays the imported model.
4. Create the endpoint.

   For more information, see Creating an Endpoint on page 70.

Importing NVIDIA NIMs from NVIDIA NGC Catalog

Import NVIDIA NIMs from NVIDIA NGC Catalog to Nutanix Enterprise AI.

Before you begin

- Ensure that you have an NGC account with an active subscription. For more information, see NVIDIA Documentation Hub.
- Certain NIM models are exclusively available with NVIDIA AI Enterprise subscription only. To import these models, ensure that you have an active NVIDIA AI Enterprise subscription. For more information, see NVIDIA Documentation Hub.
- Generate an NVIDIA NGC Personal Key and add it to Nutanix Enterprise AI. For more information, see Adding an NVIDIA NGC Personal Key on page 46.

  > **Note:** After you generate an NVIDIA NGC Personal Key, add *NGC Catalog* services to the personal key to ensure that the NIM can be imported to Nutanix Enterprise AI. For more information, see the *Personal API Key* section in the *NGC User Guide* listed in NVIDIA Documentation Hub.

About this task

To import a NIM from the NGC catalog, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Models**.
   The system displays the **Models** page.
3. Click **Import Models** > **From NVIDIA NGC Catalog**.
   The **Import Model - NVIDIA NGC Catalog** page opens, displaying all the NIMs that can run on Nutanix Enterprise AI.

   > **Note:** The NIMs validated by Nutanix display a green checkmark. For more information, see Pre-validated Models on page 51.
4. (Optional) In the **Search by Model Name** field, type the LLM name to search for an LLM.
   The system lists the matching model names as you type.
5. (Optional) To filter and view the NIM models based on the model type, select the model type from the **All Models** dropdown menu.
6. (Optional) To filter and view only the pre-validated NIM models, enable the **Show only Pre-validated Models** toggle.

7. Select a NIM and click **Import**.

   To go to the NVIDIA repository of the NIM, click the name of the NIM.

   > **Note:** If you did not add the NVIDIA NGC Personal Key to Nutanix Enterprise AI, the system prompts you to do so. Add the personal key to import the NIM. For more information, see Adding an NVIDIA NGC Personal Key on page 46.

   The **Import Model** dialog box opens.

8. In the **Model Instance Name** field, enter a name for the NIM.

   Nutanix recommends that you use the actual name of the NIM, suffixed with an identifier that is meaningful to you.

9. Click **Import**.
   The imported NIM is displayed on the **Models** page.

What to do next

After you initiate an import, you can view the status of the import on the **Models** page. The system displays one of the following states for the import operation:

- **Ready**: The NIM is imported and ready to use.
- **Failed**: The import failed due to deactivated NVIDIA NGC Personal Key, or incorrect key value added in Nutanix Enterprise AI.
- **Pending**: The system is waiting for the resources required to save the NIM.

  For example, if the required storage space is not available, Nutanix Enterprise AI maintains a **Pending** status until the storage space becomes available.
- **Processing**: The system is importing the NIM from NVIDIA NGC Catalog.

After you import a NIM to Nutanix Enterprise AI, you can deploy the NIM to an AI inference endpoint. For more information, see Creating an Endpoint on page 70. Complete the following steps:

1. Select the model.
2. Click **Actions** > **Create Endpoint** .

   The **Create Endpoint** action is enabled only for active models and the models you import. You cannot deploy models created by other users.
3. The **Create Endpoint** screen is displayed and the **Model Instance Name** field displays the imported model.
4. Create the endpoint.

   For more information, see Creating an Endpoint on page 70.

Deleting a Large Language Model

Delete an LLM that you imported to Nutanix Enterprise AI.

About this task

To delete an LLM, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.

2. From the left navigation pane, select **Models**.
   The system displays the **Models** page.

3. Select an LLM, and from the **Actions** dropdown menu, click **Delete**.
   The system prompts you to confirm the delete action.

4. In the field provided, type **delete** and click **Delete Model**.
   The selected LLM is deleted from Nutanix Enterprise AI and is no longer displayed on the **Models** page.

Calculating a Large Language Model Size

Calculate the size of a Hugging Face LLM imported to Nutanix Enterprise AI.

About this task

Before you import an LLM using the Importing a Large Language Model from Hugging Face using Model URL or ID on page 58 method, you must calculate the size of the LLM to automatically provision storage for the LLM. Knowing the model size helps you avoid deployment failures if the cluster cannot meet the resource requirements.

Procedure

1. Clone the LLM from the Hugging Face model hub to your local repository.
   The system creates a folder for the LLM in your working directory.

2. Navigate to the folder that contains the LLM.

3. Check the size of the cloned folder.
   The size of the cloned folder is the size of the LLM.

What to do next

Use this value as the model size when you import a Hugging Face LLM to Nutanix Enterprise AI, provided that the file system on your local machine matches the network file system used for storage when installing Nutanix Enterprise AI.

Configuring Proxy Server

Admins can configure a proxy server as an intermediate between Nutanix Enterprise AI and the internet to download models from Hugging Face Model Hub.

About this task

To add a proxy server in Nutanix Enterprise AI, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.

2. From the left navigation bar, click **Settings**.
   The **Third Party Credentials** tab is displayed.

3. Select the **HTTP Proxy** tab.
   The **Add Proxy Server** dialog box is displayed.

4. In the **Name** field, enter the name of the proxy server.

5. In the **Proxy Address** field, enter the IP address of the proxy server.

6. In the **Port** field, enter the port number of the proxy server.

7. (Optional) In the **Username** field, enter the username to access the proxy

8. (Optional) In the **Password**, enter the password to access the proxy.
9. In the **Protocols** section, specify either **HTTP** or **HTTPS** or both.
10. Click **Save**.
    After the proxy server is updated successfully, the proxy server is displayed in the **HTTP Proxy** tab.

ENDPOINTS IN NUTANIX ENTERPRISE AI

Nutanix Enterprise AI allows you to create AI inference endpoints.

After you successfully create an endpoint, you can deploy and use a large language model (LLM) in real-time applications. AI inference endpoints allow an application to send input data to the LLM and receive predictions or responses. The endpoint handles requests in real time, making it possible to use the LLM for tasks like generating text, answering questions, or making predictions. You can share an endpoint with developers to incorporate into their AI applications, such as chatbots or document summarizations.

> **Important:** By default, Nutanix Enterprise AI does not limit or modify the maximum number of inference requests that an endpoint can handle simultaneously.

Perform the following actions to manage an endpoint:

- Create an endpoint
- Delete an endpoint
- Test an endpoint
- Hibernating an Endpoint on page 76
- Resuming an Endpoint on page 76

Viewing Endpoints in Nutanix Enterprise AI

The **Endpoints** page displays all the endpoints that you created in Nutanix Enterprise AI.

About this task

To view the endpoints, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints. For information on the endpoints attributes displayed in the page, see Endpoints Attributes on page 66.
3. (Optional) To filter endpoints, follow these steps:
   a. Click **Modify Filters**.
      The **Filters** pane opens.
   b. Select one of the following filters:
      » Hibernated
      » Non-Hibernated

Endpoints Attributes

The following table describes the endpoint attributes that appear on the **List** page.

Table 19: Endpoints Attributes - Description

Attributes	Description	Values
Name	Displays the user-provided name when creating the endpoint.	Endpoint name
Model	Displays the model name you specified when you imported the model. It also displays the type of the imported model.	Model name and model type
Accelerator	Displays the accelerator type and count.	• Displays the number and name of the GPUs per instance if the accelerator is GPU. • Displays INTEL AMX vCPU if the accelerator is an AMX enabled intel CPU. • Displays CPU if the processor does not support AMX.
Number of Instances	Displays the number of Kubernetes pods configured to serve the endpoint.	Number of Instances
Created By	Displays the name of the user who created the endpoint.	User name
Status	Displays the current status of the endpoint.	Active, Failed, Hibernated, Pending, Processing

Viewing the Endpoint Details

View detailed information about endpoints.

About this task

To view the detailed information about an endpoint, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints that you created.
3. Click an endpoint.
   The details page opens, displaying detailed information about the endpoint in widgets.
4. Click **Overview**.

   For information on the widgets displayed in the page, see Table 20: Widgets in the Overview tab on page 68.
5. Click **Metrics** > **Usage**.

   For information on the widgets displayed in the page, see Table 21: Metrics tab Widgets - Description on page 69.

Widgets in the Overview tab of an Endpoint Details Page

The following table describes the widgets that appear on the endpoint details page.

Table 20: Widgets in the Overview tab

Widget Name	Information Provided
Details	Provides the following: • **Endpoint Name**: Displays the user-provided name of the endpoint. • **Description**: Displays the user-provided description of the endpoint. • **Status**: Displays the current status of the endpoint. • **Model Instance Name**: Displays the user-provided name of the large language model (LLM) on which the endpoint is deployed. • **Model Type**: Displays the type of imported LLM model on which the endpoint is deployed. • **Inference Engine**: Displays the type of inference engine running the LLM model. • **Accelerator**: Displays the name of the GPU card used to run the model.
Endpoint Access	Provides the following: • **Endpoint URL**: Displays the URL of the endpoint. • **API Keys**: Displays the API key required to access the endpoint. The key is masked. Hover over the tooltip to go to the **API Keys** page. You can create new API keys and manage existing keys from this page. For more information, see Viewing API Keys in Nutanix Enterprise AI on page 78. • **View Sample Request**: Click this option to view the sample code that contains the URL and API key required to access the endpoint. For more information, see Viewing Sample API Code on page 70.

Widget Name	Information Provided
Number of Compute Instances	Displays the: • Maximum number of instances configured to serve the endpoint. • Number of instances actively running to serve the endpoint. • Number of vCPUs and the memory usage per instance actively in use to serve the endpoint.
Requests Summary	Displays the total number of API requests received by the endpoint.
Token Usage	Displays the number of tokens received and generated by this endpoint.

Widgets in the Metrics tab

The following table describes the widgets that appear in the **Metrics** tab on the endpoint details page. You can select **Usage** or **Performance** from the drop-down menu to view the required metrics

- Metrics will be available on the dashboard based on the capability of the Inference Engine.
- Metrics may display the following state:
  - Not applicable when a model using this endpoint does not support a metric.
  - No Data available when there is no data for a specific interval.
- The graph is displayed for a specified `interval` you select from the dropdown menu on the right of the widget. The `interval` involves **Last 15 minutes**, **Last 1 hour**, and **Last 24 hours**.

Table 21: Metrics tab Widgets - Description

Option	Widget Name	Information Provided
Usage metrics	Requests Trend	• Displays the total number of API requests received by the endpoint. The API requests for all API keys are displayed by default. You can filter the metrics for an API key using the drop-down menu.
	Token Usage	Displays the number of tokens processed and generated by this endpoint.
Performance metrics	Latency	Displays the graph of the time taken by an endpoint to respond to an API request.
	Number of Requests waiting at the Inference Engine	Displays the graph of the requests that have reached the endpoint but are in queue.

Option	Widget Name	Information Provided
	Number of Requests running at the Inference Engine	Displays the graph of requests that are currently being processed by that endpoint.
	Time to First Token (TTFT)	Displays the graph of the time interval between when a request reaches the inference endpoint and when the first generated token is streamed back to the client.
	Time per Output Token (TPOT)	Displays the graph of the average time that the endpoint takes to generate each output token after sending the first one
	Output Tokens per Second	Displays the graph of the token throughput of the inference endpoint.

Viewing Sample API Code

The inference endpoints in Nutanix Enterprise AI can be accessed using a URL and an API Key. Share the sample code with the developers to access and integrate endpoints into their applications.

About this task

To view the sample code that contains the URL and API key required to access the endpoint, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints that you created.
3. Click an endpoint.
   The details page opens, displaying detailed information about the endpoint in widgets.
4. From the **Endpoints Access** widget, click **View Sample Request**.
   The **Endpoints Access** dialog box displays the sample code. The sample code contains the URL required to access the endpoint.
5. (Optional) To perform tasks that require the endpoint to interact with an external tool, select the **Enable Tool Calling** checkbox.

   For example, enable tool calling to check the current weather at a location.
6. (Optional) To copy the sample code, click **Copy Code**.

   You can share the code with the developers to access the endpoints and integrate them into their applications.

Creating an Endpoint

Create an inference endpoint in Nutanix Enterprise AI and share it with developers to incorporate into their AI applications. The inference endpoint accepts requests and sends back responses.

Before you begin

Import a model to Nutanix Enterprise AI, and ensure that the import status is **Active**. For more information, see Importing a Large Language Model from Hugging Face on page 56, Importing NVIDIA NIMs from NVIDIA NGC Catalog on page 62, or Importing a Large Language Model Manually from Hugging Face on page 60.

About this task

To create an endpoint from an LLM, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints that you created.
3. Click **Create Endpoint**.
   The system displays the **Basics** tab of the **Create an Endpoint** dialog box.
4. In the **Endpoint Name** field, enter a name for the endpoint.

   Nutanix recommends that you provide a name that is meaningful and identifiable to you.

   > **Note:** The name must meet the following complexity requirements:
   >
   > - Must use only lowercase letters, numbers, or special characters (-).
   > - Must start with a lowercase letter.

5. (Optional) In the **Description** field, enter a description for the endpoint.
6. From the **Model Type** dropdown menu, select the type of model required to deploy the endpoint.
7. From the **Model Instance Name** dropdown menu, select an active imported model required to deploy the endpoint.

8. Specify the acceleration type. Do any of following:

    - Enable GPU acceleration. Complete the following steps:

        1. From the **Acceleration Type** dropdown menu, select **GPU Passthrough**.

            > **Note:** The **Acceleration Type** drop-down menu displays **GPU Passthrough** only if the cluster has GPU nodes and the model you selected in Step 7 supports GPU execution.

        2. From the **GPU Card** dropdown menu, select the required GPU card.

            For information on the GPU cards supported for a model, see GPU Requirements in Nutanix Enterprise AI Requirements.

    - Enable CPU acceleration. Complete the following steps:

        1. From the **Acceleration Type** dropdown menu, select **CPU**.

            > **Note:** The **Acceleration Type** dropdown menu displays **CPU** only if the model you selected in Step 7 supports CPU execution.

        2. (Optional) If you have Intel Xeon 4th-gen or newer CPUs , select the **Optimise endpoint to run on Intel® AMX enabled CPUs** checkbox.

            - This checkbox is selected by default if all the CPUs in the cluster are Intel® Advanced Matrix Extensions (AMX) enabled CPUs

            - This checkbox is grayed out by default if none of the CPUs in the cluster are Intel® AMX enabled CPUs

    The built-in accelerator available in Intel® Xeon® 4th-gen (or newer) CPUs is enabled.

9. Specify the API key to access endpoints. Do one of the following:

    » If you did not create an API key to access the endpoint, click **Create a New API Key** to create a new API key.

    For more information, see Creating an API Key on page 79.

    » If you already created an API key, select an active API key from the **API Keys** dropdown menu to access the endpoint.

10. Click **Next**.

    The **Configuration** tab is displayed.

11. From the **Inference Engine** dropdown menu, select the type of inference engine required to deploy the endpoint.

12. In the **Number of GPUs (Per Instance)** field, enter the number of GPUs per instance required to run the model.

    > **Important:** This field is displayed only if you have selected the **Acceleration Type** as **GPU Passthrough**. The minimum number of GPUs 1. The supported numbers are 1,2,4 and 8.

13. In the **Number of Instances** field, enter the number of instances required to serve the endpoint.

14. (Optional) Select the **Use Custom Configuration for Instances** checkbox to manually configure the inference engine type, number of vCPUs, and memory.

    The checkbox is selected in case of a non-validated model. In other cases, this checkbox remains clear because the system automatically configures the inference engine type, number of vCPUs, and memory according to the model you select in Step 7 and your cluster configuration, and displays the values in the respective fields.

15. (Optional) In the **vCPUs (Per Instance)** field, enter the number of Kubernetes node pool vCPUs to assign to the endpoint.

    > **Note:** Ensure that you assign the vCPUs according to your cluster configuration. If you assign a value that exceeds the number of vCPUs configured in the cluster, the system displays an error.

16. (Optional) In the **Memory (Per Instance)** field, enter the amount of memory to assign to the endpoint. This field is auto-populated based on the selected LLM and displays the recommended memory as per your cluster configuration.

    > **Note:** Ensure that you assign the memory according to your cluster configuration. If you assign a value that exceeds the memory configured in the cluster, the system displays an error.

17. In the **Context Length** field, enter the maximum number of tokens that can be processed based on the selected pre-validated Hugging Face model, the CPU or GPU card, and the number of GPUs.

    This limit includes both input and output tokens. You can also manually configure this field based on your available system resources, including the Hugging Face model, the CPU or GPU card, and the number of GPUs.

    > **Note:**
    >
    > - The context length determines the combined length of the input text and the generated output.
    > - This field appears only if you select a Hugging Face model in Step 7 and the **Inference Engine** is **vLLM** or **TGI (Text Generation Inference)**.
    > - The system displays a default value only for pre-validated models.
    > - The maximum context length for a custom model with GPUs enabled is 1048576, and without GPUs, it is 4096. If you enter a higher value, the system displays an error.

18. (Optional) Click **Back**.

    The **Basics** tab is displayed. The values you specify in the **Number of GPUs (Per Instance)**, **vCPUs (Per Instance)**, **Memory (Per Instance)** and the **Context Length** field are not saved. Therefore, complete the following steps:

    a. Click **Next**.
    b. Update the **Number of GPUs (Per Instance)** field.
    c. Update the **vCPUs (Per Instance)** field.
    d. Update the **Memory (Per Instance)** field.
    e. Update the **Context Length** field.

19. Click **Next**.

    The **Summary** tab is displayed.

20. Click **Create**.
    The system creates the inference endpoint, which is displayed on the **List** page.

What to do next

After you create an inference endpoint, you can view the status of the endpoint on the **Endpoints** page. The system displays one of the following states for the operation:

- **Active**: The endpoint is created and ready to use.

  > **Note:** An inference endpoint accepts requests and sends back responses only if the status of the endpoint displays **Active**.

- **Failed**: Endpoint creation failed due to unauthorized NVIDIA NGC Personal Key, or *NGC Catalog* services access restriction for the personal key.

  > **Note:** If the system displays an error message other than the ones listed, run the following command, and contact Nutanix Support with the endpoint specifications provided in the output.
  >
  > ```
  > $ kubectl describe isvc endpoint_name -n nai-admin
  > ```
  >
  > Replace `endpoint_name` with the name of the endpoint on the **List** page.

- **Pending**: The system is waiting for the resources required to create the endpoint.

  For example, if the required number of GPUs are not available, Nutanix Enterprise AI maintains a **Pending** status until the GPUs becomes available.

- **Processing**: The system is creating the endpoint.

After you successfully create an endpoint, you can test it to ensure that it is configured correctly. For more information, see Testing an Endpoint on page 74.

Deleting an Endpoint

Delete an endpoint in Nutanix Enterprise AI.

About this task

To delete an endpoint, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints that you created.
3. Select the checkbox associated with an endpoint, and from the **Actions** dropdown menu, click **Delete**.
   The system prompts you to confirm the delete action.
4. In the field provided, type **delete** and click **Delete Endpoint**.
   The endpoint is deleted from Nutanix Enterprise AI and is no longer displayed on the **List** page.

Testing an Endpoint

Test an endpoint created in Nutanix Enterprise AI using a text generation LLM to verify the endpoint configuration and to ensure that the API connection to an LLM is active.

Before you begin

> **Note:** To test an endpoint created using non-text generation LLMs, you must copy the sample code that contains the URL and API key required to access the endpoint and run it in your application. For more information, see Viewing Sample API Code on page 70.

Create an inference endpoint, and ensure that the status of the endpoint is **Active**. For more information, see Creating an Endpoint on page 70.

About this task

To test an endpoint, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints that you created.
3. Click the name of an endpoint to test.
   The endpoint details page opens, displaying detailed information about the endpoint in widgets.
4. Click **Test**.

   Test functionality is not supported for Image Generation and Image Classification endpoints.

   The system displays the **Test Endpoint** dialog box.
5. Perform one of the following:

   » Select **Sample Request**, choose a default question, and click **Test**.

   » Select **Custom Request**, enter a question in the field provided, and click **Test**.

   If the endpoint is configured correctly, the system displays the **Status** as **Succeeded** and the **Result** text box displays a generated answer.
6. To close the dialog box and return to the endpoint details page, click **Done**.

Accessing an Endpoint using Open AI Compatible Clients

Access an endpoint created in Nutanix Enterprise AI using any OpenAI compatible clients.

Before you begin

Ensure that the following requirements are met before you access an endpoint created in Nutanix Enterprise AI using any OpenAI compatible clients.

- The AI/ML Admin user has created the endpoints in Nutanix Enterprise AI. For more information, see Creating an Endpoint.
- The AI/ML Admin user has created an API key in Nutanix Enterprise AI and attached it to the endpoints. For more information, see Creating an API Key.
- The API key created in Nutanix Enterprise AI is made available for your use.

About this task

Nutanix Enterprise AI supports the following OpenAI compatible endpoints:

- /v1/images/generations

- /v1/chat/completions
- /v1/models
- v1/embeddings

Procedure

- For information on how to access an endpoint using an OpenAI-compatible client, see the client-specific documentation.

Endpoint Hibernation

Hibernating an endpoint in Nutanix Enterprise AI pauses the activity of the endpoint and releases the associated compute resources without deleting the endpoint.

Hibernating an endpoint optimizes resource usage while preserving endpoint configuration for future use. You can hibernate an endpoint when the cluster that hosts Nutanix Enterprise AI has limited available resources. You can resume a hibernated endpoint after you add more compute resources to the cluster or if the cluster already has enough resources.

Hibernating an Endpoint

Hibernate an existing endpoint in Nutanix Enterprise AI.

About this task

To hibernate an endpoint, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI
2. From the left navigation pane, select **Endpoints**.
   The List page opens, displaying a summary of all the endpoints.
3. Select the checkbox associated with an endpoint, and from the **Actions** dropdown menu, click **Hibernate**.
   The system prompts you to confirm the hibernation action.
4. In the field provided, type **hibernate** and click **Hibernate**.
   The system hibernates the endpoint and displays the endpoint status as **Hibernated** on the List page.

Resuming an Endpoint

Resume a hibernated endpoint in Nutanix Enterprise AI.

Before you begin

- Ensure that the cluster that hosts Nutanix Enterprise AI has enough resources to meet the endpoint requirements.
- If the cluster does not have enough resources, allow a five-minute gap between the Hibernate and Resume actions to ensure that Kubernetes fully releases the resources used by the hibernated endpoint.
- If the cluster has enough resources to meet the endpoint requirements, you can resume the endpoint immediately. However, the startup time might increase depending on the resource usage of the endpoint.

About this task

To resume an endpoint, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI
2. From the left navigation pane, select **Endpoints**.
   The List page opens, displaying a summary of all the endpoints.
3. Select the checkbox associated with an endpoint, and from the **Actions** dropdown menu, click **Resume**.
   The system prompts you to confirm the resume action.
4. Click **Resume**.
   The system resumes the endpoint.

   > **Note:** After you click **Resume**, the system provisions the compute resources required to run the endpoint. If the cluster has the required resources, the system displays the endpoint status as **Ready** on the List page. If the cluster does not have the required resources, the system displays the endpoint status as **Pending** until the resources are available.

API KEYS IN NUTANIX ENTERPRISE AI

API keys are required to access the endpoints to integrate with the application.

Perform the following actions to manage an API key:

- Create an API key
- Update an API key
- Delete an API key
- Activate or deactivate an API key

Viewing API Keys in Nutanix Enterprise AI

The **API Keys** page displays all the API keys that you created in Nutanix Enterprise AI.

About this task

To view the API keys, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints that you created.
3. Click **API Keys**.
   The **API Keys** page opens, displaying a summary of all the API keys that you created. For information on the attributes of the API keys displayed in the page, see API Keys Attributes on page 78.

API Keys Attributes

The following table describes the API key attributes that appear on the **API Keys** page.

Table 22: API Keys Attributes - Description

Field	Description	Values
Key Name	Displays the user-provided name when creating the key.	Key name
Key Value	Displays the API key created by the user. The key is masked.	Key
Endpoints	Displays the name of the endpoint associated with the API key.	Endpoint name
Created By	Displays the name of the user who created the key.	User name
Status	Displays the current status of the API key.	Active, Inactive

Creating an API Key

Create an API key in Nutanix Enterprise AI to allow an application to access an inference endpoint.

About this task

To create an API key, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints that you created.
3. Click **API Keys**.
   The **API Keys** page opens, displaying a summary of all the API keys that you created.
4. Click **Create a New Key**.
   The **Create API Key** dialog box opens.
5. In the **Key Name** field, enter a name for the API key.

   > **Note:** Nutanix recommends that you provide a name that is meaningful and identifiable to you.

6. To assign the key to an endpoint, from the **Endpoints** dropdown menu, select the endpoint.
7. Click **Create**.
   The system creates the API key and opens the **API Key Details** dialog box that displays the details of the API key.
8. Click **Copy Key**.

   > **Important:** Ensure that you copy the key and store it securely. You cannot view the key again after you close the **API Key Details** dialog box. If you lose the key, you must generate a new one.

Activating or Deactivating an API Key

Activate or deactivate an API key that you created in Nutanix Enterprise AI.

About this task

To activate or deactivate an API key, follow these steps:

> **Note:** By default, an API key is activated when you create it.

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints that you created.
3. Click **API Keys**.
   The **API Keys** page opens, displaying a summary of all the API keys that you created.

4. Perform one of the following:

    » To activate an inactive API key, select the checkbox associated with the key, and from the **Actions** dropdown menu, click **Activate**.

    » To deactivate an active API key, select the checkbox associated with the key, and from the **Actions** dropdown menu, click **Deactivate**.

    The system prompts you to confirm the action.

5. Click **Confirm**.
    The system displays the new status of the key.

Updating an API Key

Update an existing API key that you created in Nutanix Enterprise AI.

About this task

To update an API key, follow these steps:

You can perform the following API key configuration:

- Assign the same API key to multiple endpoints.
- Remove an endpoint to which the API key is assigned.

Procedure

1. Log in to Nutanix Enterprise AI.

2. From the left navigation pane, select **Endpoints**.
    The **List** page opens, displaying a summary of all the endpoints that you created.

3. Click **API Keys**.
    The **API Keys** page opens, displaying a summary of all the API keys that you created.

4. Select the checkbox associated with a key, and from the **Actions** dropdown menu, click **Update**.
    The **Update API Key** dialog box opens.

5. Perform one of the following actions:

    » To assign the API key to an endpoint, from the **Endpoints** dropdown menu, select the checkbox associated with the endpoint.

    » To unassign the API key from an endpoint, from the **Endpoints** dropdown menu, clear the checkbox associated with the endpoint.

    > **Note:** You cannot change the key name or the key.

6. Click **Update**.

Deleting an API Key

Delete an existing API key that you created in Nutanix Enterprise AI.

About this task

To delete an API key, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Endpoints**.
   The **List** page opens, displaying a summary of all the endpoints that you created.
3. Click **API Keys**.
   The **API Keys** page opens, displaying a summary of all the API keys that you created.
4. Select the checkbox associated with an API key, and from the **Actions** dropdown menu, click **Delete**.
   The system prompts you to confirm the delete action.
5. In the field provided, type **delete** and click **Delete API Key**.
   The API key is deleted from Nutanix Enterprise AI and is no longer displayed on the **API Keys** page.

VIEWING AUDIT EVENTS IN NUTANIX ENTERPRISE AI

An AI/ML Admin user can view a list of all the event messages audited and captured as part of the audit logs on the **Audit Events** page of Nutanix Enterprise AI.

About this task

Event messages describe actions such as importing a model, resetting the password, a user logging in and logging out, updating or deleting an endpoint, or creating an endpoint, API key, or user. Unlike alerts, event messages are simply informational without the need to acknowledge or resolve.

> **Note:** By default, no retention period exists for the event messages.

To view the event messages, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Audit Events**.

   The **Audit Events** page opens, displaying a list of all the event messages that are audited and captured as part of the audit logs. For information on the fields that appear on the page, see Audit Events Summary on page 82.

Audit Events Summary

An AI/ML Admin user can view the list of event messages in Nutanix Enterprise AI.

The following table describes the fields that appear on the **Audit Events** page.

Table 23: Audit Events Summary - Field Description

Field	Description	Values
Description	Displays the name of the event.	Event name
Action	Displays the type of operation that took place. The possible operation types depend on the entity type.	Create, Delete, Log In, Log Out, Reset Password, Update, Expire, Pulse Upload Event
Entity Type	Displays the entity type, such as an endpoint, cluster, model, and so on.	API Key, Cluster Config, Endpoint, Hugging Face Token, Model, NVIDIA NGC API Key, License, User
User	Displays the name of the user who performed the event.	User name
Time	Displays the date and time when the event occurred.	Date and time

Audit Events Filters

The following table describes the filters that are available on the **Audit Events** page. For information on how to filter the list of event messages, see Filtering Audit Events on page 83.

Table 24: Audit Events Filters - Field Description

Filter	Description	Values
Entity Type	Select the checkboxes of one or more entities to filter for actions on those entity types.	API Key, Cluster Config, Endpoint, Hugging Face Token, Model, NVIDIA NGC API Key, License, User
Action	Select the checkboxes of one or more actions to filter for those actions on an entity.	Create, Delete, Log In, Log Out, Reset Password, Update, Expire, Pulse Upload Event
User	Enter a user name and press Enter to filter for actions requested by that user. Click **+ Add Rule** to create a new rule.	User name

Filtering Audit Events

An AI/ML Admin user can filter the list of event messages in Nutanix Enterprise AI.

About this task

To filter the event messages, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Audit Events**.

   The **Audit Events** page opens, displaying a list of all the event messages that are audited and captured as part of the audit logs.
3. Click **Modify Filters**.

   The **Filters** pane opens, displaying the available filters. For information on the available filters, see Table 24: Audit Events Filters - Field Description on page 83.
4. To view the event messages based on a filter, select the checkbox associated with the filter.

   > **Note:** You can apply multiple filters together. For example, if you want to filter for endpoint creation, you can select the **Endpoint** checkbox in **ENTITY TYPE** and then select the **Create** checkbox in **Action**.
5. To export the filtered event messages, follow these steps:

   a. Click **Export**.

   b. Select **All entities** or **Only Filtered Entities**.

   c. Click **Export**.

USER ROLES IN NUTANIX ENTERPRISE AI

After you deploy Nutanix Enterprise AI at your site, you can create the following roles: **AI/ML Admin** and **AI/ML User**.

The following table lists the privileges associated with the super admin, AI/ML admin, and AI/ML user roles.

> **Note:** The first AI/ML admin user who installs Nutanix Enterprise AI on the Kubernetes cluster is called a super admin.

Table 25: User Role Privileges

Privileges	Super Admin	AI/ML Admin	AI/ML User
Add a user	Yes	Yes	No
Update the details of a user	Yes	Yes	No
Activate or deactivate a user	Yes	Yes	No
Reset the password of a user	Yes	Yes	No
Change the language settings	Yes	Yes	No
Adding a license	Yes	Yes	No
View the summary of the resources available in the Kubernetes cluster that hosts Nutanix Enterprise AI	Yes	Yes	Yes
View the end user license agreement (EULA) page	Yes	No	No
View the accept telemetry installation workflow page	Yes	No	No
View the large language models (LLMs) imported by a user.	Yes	Yes	Yes **Note:** An AI/ML user can view only the LLMs imported by the user.
Import an LLM	Yes	Yes	Yes
Delete an imported LLM	Yes	Yes	Yes **Note:** An AI/ML user can delete only the LLMs imported by the user.

Privileges	Super Admin	AI/ML Admin	AI/ML User
View the endpoints created by a user	Yes	Yes	Yes **Note:** An AI/ML user can view only the endpoints created by the user.
Create an endpoint	Yes	Yes	Yes
Delete an endpoint	Yes	Yes	Yes **Note:** An AI/ML user can delete only the endpoints created by the user.
Update an endpoint	Yes	Yes	Yes **Note:** An AI/ML user can update only the endpoints created by the user.
Test an endpoint	Yes	Yes	Yes **Note:** An AI/ML user can test only the endpoints created by the user.
View the API key names created by a user	Yes	Yes	Yes **Note:** An AI/ML user can view only the API key names created by the user.
Create an API key	Yes	Yes	Yes
Update an API key	Yes	Yes	Yes **Note:** An AI/ML user can update only the API keys created by the user.
Activate or deactivate an API key	Yes	Yes	Yes **Note:** An AI/ML user can activate or deactivate only the API keys created by the user.

Privileges	Super Admin	AI/ML Admin	AI/ML User
Delete an API key	Yes	Yes	Yes **Note:** An AI/ML user can delete only the API keys created by the user.
Add or update third-party access tokens	Yes **Note:** A Super Admin user can add or update only the tokens created by the user.	Yes **Note:** An AI/ML Admin user can add or update only the tokens created by the user.	Yes **Note:** An AI/ML user can add or update only the tokens created by the user.
View Audit Events	Yes	Yes	No
Enable or Disable Pulse	Yes	Yes	No
Update Model Access Control Policy	Yes	Yes	No
Add or update HTTP Proxy settings	Yes	Yes	No

Viewing User Roles in Nutanix Enterprise AI

View all the user roles that you created in Nutanix Enterprise AI.

About this task

To access the **Users** page, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Admin Configuration** > **Users**.
   The **Users** page opens, displaying a list of users you created.

Viewing User Role Details

View the user role details provided when you created the user. You can share these details with the user so that they can access Nutanix Enterprise AI.

About this task

To view the user role details, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Admin Configuration** > **Users**.
   The **Users** page opens, displaying a list of users you created.
3. Click the name of a user.
   The **User Details** dialog box opens and displays the details you provided when you created the user.

4. (Optional) Click **Copy to Clipboard**.

   Copy and share the details with the users so that they can access Nutanix Enterprise AI.

Creating Users in Nutanix Enterprise AI

A user with the AI/ML admin role can create users with the AI/ML admin and AI/ML user roles.

About this task

To create a new user, follow these steps.

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Admin Configuration** > **Users**.
   The **Users** page opens.
3. Click **Add User**.
   The **Add User** dialog box opens.
4. Perform the following in the indicated fields:
   a. **Role**: Select the role to assign to the user from the dropdown menu.
   b. **Email Address**: Enter the email address of the user.
   c. **First Name**: Enter the first name of the user.
   d. **Last Name**: Enter the last name of the user.
   e. **Username**: Enter the username to log in to Nutanix Enterprise AI.
   f. **Password**: Enter the password to log in to Nutanix Enterprise AI.
5. Click **Add**.
   The system creates the new user and displays it on the **Users** page.

   Nutanix Enterprise AI does not automatically share the user credentials with the new user. You must manually send them to the user.

Updating Users in Nutanix Enterprise AI

An AI/ML admin user can update the details of an existing user in Nutanix Enterprise AI.

About this task

To update the details of an existing user, perform the following steps.

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Admin Configuration** > **Users**.
   The **Users** page opens, displaying a list of users you created.
3. Select the checkbox associated with the user, and from the **Actions** dropdown menu, click **Update**.
   The **Update User** dialog box opens.

4. Make the required changes and click **Update**.
The system updates the user details and displays them on the **Users** page.

Activating or Deactivating Users in Nutanix Enterprise AI

If you are an AI/ML admin user, you can activate or deactivate a user you created in Nutanix Enterprise AI.

About this task

To activate or deactivate a user, follow these steps:

> **Note:** By default, a user is activated when you create the user. When you deactivate a user, the user cannot log in to Nutanix Enterprise AI and deploy models. However, the resources created by the user continue to exist.

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Admin Configuration** > **Users**.
The **Users** page opens, displaying a list of users you created.
3. Perform one of the following:
    - To activate an inactive user, select the checkbox associated with the user and from the **Actions** dropdown menu, click **Activate**.
    - To deactivate an active user, select the checkbox associated with the user and from the **Actions** dropdown menu, click **Deactivate**.

    The system prompts you to confirm the action.
4. In the confirmatory dialog box that opens, perform one of the following based on your choice in Step 3:
    - Click **Activate** to activate the user.
    - Click **De-Activate** to deactivate the user.

    The system displays the updated status of the user on the **Users** page.

Resetting User Passwords in Nutanix Enterprise AI

An AI/ML admin user can reset the password of other users in Nutanix Enterprise AI.

About this task

To reset the password of an existing user, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Admin Configuration** > **Users**.
The **Users** page opens, displaying a list of users you created.
3. Select the checkbox associated with the user, and from the **Actions** dropdown menu, click **Reset Password**.
The **Reset Password** dialog box opens.
4. Enter the new password for the user and click **Reset Password**.
The system resets the existing password and assigns a new password to the user.

VIEWING KUBERNETES CLUSTER RESOURCES AND HEALTH METRICS

View the resources available in the Kubernetes cluster that hosts Nutanix Enterprise AI.

About this task

To view the Kubernetes cluster resources, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, select **Infrastructure**.
   The **Overview** tab in the **Infrastructure** page opens, displaying a summary of the resources available in the Kubernetes cluster. For information on the fields that appear on the page, see Kubernetes Cluster Summary on page 89.

   For information on the health and infrastructure usage statistics of the cluster and the nodes, see Viewing Infrastructure Usage Metrics on page 91.

Kubernetes Cluster Summary

The following table describes the fields that appear on the **Kubernetes Cluster** page.

Table 26: Kubernetes Cluster Summary - Field Description

Field	Description

Name	• Displays the name of the node in the cluster
	• Displays the health of the node.
	  • Green dot indicates node is online
	  • Grey dot indicates node is offline.

Version	Displays the Kubernetes version running on the node pool.
vCPU	Displays the number of vCPUs assigned to the node pool.
Memory	Displays the memory available to the node.
Disk	Displays the storage space available to the node.
Accelerators	Displays the name and number of GPUs assigned to this node when you hover over the hover info icon.

Field	Description
GPU Memory	Displays the total GPU memory available on the node. **Important:** The number of GPUs displayed might be incorrect if the GPU operator pods in your Kubernetes cluster are unhealthy. To fix these unhealthy pods, contact NVIDIA support.

VIEWING INFRASTRUCTURE USAGE METRICS

View the infrastructure usage metrics of the Kubernetes cluster that hosts Nutanix Enterprise AI and of the individual nodes in the cluster.

About this task

To view the infrastructure usage metrics, follow these steps:

Procedure

1. Log in to Nutanix Enterprise AI.
2. From the left navigation pane, do one of the following:

    - Select **Dashboard**.

      The system displays the dashboard with the widgets.

      From the **Infrastructure Summary** widget, click **View Details**.
    - Select **Infrastructure** > **Usage**.

    The system displays the **Usage** tab in the **Infrastructure** page.

    For information on the cluster usage metricsd, see Cluster Usage Metrics on page 91 and for node usage statistics, see Node Usage Metrics on page 92.

Cluster Usage Metrics

The following table describes the metrics that appear in the **Infrastructure** page when you select the **Usage** tab and then select **Clusters** from the drop-down menu on the left.

- The graphs displays No Data when there is no data for a specific interval.
- The graph is displayed for a specified `interval` you select from the dropdown menu on the right of the widget. The `interval` involves **Last 15 minutes**, **Last 1 hour**, and **Last 24 hours**.

Table 27: Cluster Usage and Health - Metrics Description

Metrics	Description
Memory Usage	Displays the graph of memory capacity currently being used by the Kubernetes cluster that hosts Nutanix Enterprise AI.
CPU Usage	Displays the graph of CPU capacity currently being used by the Kubernetes cluster that Nutanix Enterprise AI.
Nodes	Displays a summary of all the nodes in the cluster that hosts Nutanix Enterprise AI, along with their names, number of GPUs, CPU capacity, and memory capacity currently in use.

Node Usage Metrics

The following table describes the metrics that appear in the **Infrastructure** page when you select the **Usage** tab and then select a node from the drop-down menu on the left.

- The graphs displays No Data when there is no data for a specific interval.
- The graph is displayed for a specified `interval` you select from the dropdown menu on the right of the widget. The `interval` involves **Last 15 minutes**, **Last 1 hour**, and **Last 24 hours**.

Table 28: Infrastructure Usage and Health - Node Metrics Description

Widget	Description
Memory Usage	Displays the graph of memory capacity used by the node.
CPU Usage	Displays the graph of CPU capacity used by the node.
GPU Utilization	Displays the graph of GPU capacity used by the node.
GPU Memory Usage	Displays the percentage of GPU memory capacity used by the node.

NUTANIX ENTERPRISE AI PULSE TELEMETRY

Pulse telemetry collects data to improve and optimize Nutanix Enterprise AI.

When you enable Pulse, you share data about your Nutanix Enterprise AI application usage with Nutanix. The information collected helps Nutanix understand how you are using Nutanix Enterprise AI to help us improve the product. For more information on the data that Pulse gathers from Nutanix Enterprise AI, see Nutanix Enterprise AI Data Shared with Nutanix on page 93.

Nutanix processes data that Pulse sends in a manner consistent with your agreement with Nutanix and, where applicable, by the Nutanix Privacy Statement. For more information, see Nutanix Privacy Statement.

You can review the Pulse settings in Nutanix Enterprise AI. For more information, see Enabling Nutanix Enterprise AI Pulse Telemetry on page 95 or Disabling Nutanix Enterprise AI Pulse Telemetry on page 95.

Nutanix Enterprise AI Data Shared with Nutanix

This topic enlists the entities and the data shared from Nutanix Enterprise AI with Nutanix through Pulse telemetry.

Some of this information might be anonymized depending on your settings. This list is not exhaustive; for more details about the Pulse telemetry your clusters send to Nutanix, contact Nutanix Support.

Table 29: Pulse Data

Entity	Data Collected
Inference endpoint	Name of the inference endpoint
	Total number of API requests sent to the inference endpoint
	Total number of API keys assigned to the inference endpoint
	Name of the imported large language models
	Number of large language models used in inference endpoints
	Number of GPUs and the type of GPU assigned to the inference endpoint
	Details of the Kubernetes cluster that hosts Nutanix Enterprise AI
	Details of the Kubernetes cluster on which Nutanix Enterprise AI is licensed
	Current status of the endpoint
	Details of the imported model used to create the endpoint
Kubernetes cluster	Kubernetes version name
	Nutanix Enterprise AI version number

Entity	Data Collected
	Total number of worker nodes present in the Kubernetes cluster
	vCPUs assigned to the Kubernetes cluster
	Disk size allotted to the Kubernetes cluster
	Memory allotted to the Kubernetes cluster
	Number of accelerators allotted to the Kubernetes cluster
	Name of the accelerator allotted to the Kubernetes cluster
	Actual memory available per accelerator
	Details of the Kubernetes cluster that hosts Nutanix Enterprise AI
	Details of the Kubernetes cluster on which Nutanix Enterprise AI is licensed
	Details of the Nutanix Enterprise AI license key applied to the cluster
	Names of the models imported to the Nutanix Enterprise AI instance
User	Total number of admin users
	Total number of users
	Total number of Hugging Face tokens, and NVIDIA NGC tokens created in the Nutanix Enterprise AI instance
	Details of the Kubernetes cluster that hosts Nutanix Enterprise AI
	Details of the Kubernetes cluster on which Nutanix Enterprise AI is licensed
	Number of API keys available in the Nutanix Enterprise AI instance
Nutanix Enterprise AI features	Details of the Kubernetes cluster that hosts Nutanix Enterprise AI
	Details of the Kubernetes cluster on which Nutanix Enterprise AI is licensed
	Number of inference requests triggered for an endpoint
	Number of API keys associated with an endpoint
	Number of input and output tokens processed by an endpoint
	Number of times an endpoint is created using a specific model type or an endpoint created using a specific model type is hibernated

Entity	Data Collected
	Number of times a model is imported using a specific import method
	Number of times a metric is queried
	Number of times the access control settings were updated
	Number of times an API for a resource is queried

Enabling Nutanix Enterprise AI Pulse Telemetry

Pulse shares data about your Nutanix Enterprise AI application usage with Nutanix. This topic describes the steps to enable Nutanix Enterprise AI Pulse telemetry.

Before you begin

Ensure that your firewall allows the IP addresses of the Kubernetes cluster that hosts Nutanix Enterprise AI because Pulse data is sent from this cluster to insights.nutanix.com over port 443 using the HTTPS REST endpoint.

About this task

To enable Nutanix Enterprise AI Pulse telemetry, follow these steps:

> **Note:**
>
> - When you log in to Nutanix Enterprise AI for the first time after an installation or upgrade, the system checks whether Pulse is enabled. If it is not enabled, a screen appears recommending you to enable Pulse. To enable Pulse, select the **Enable Pulse** checkbox, and click **Save and Proceed**. To continue without enabling Pulse in Nutanix Enterprise AI, clear the **Enable Pulse** checkbox and click **Save and Proceed**.
> - Nutanix does not collect any personally identifiable information (PII) through Pulse.

Procedure

1. Log in to Nutanix Enterprise AI as an AI/ML Admin user.
2. From the left navigation bar, click **Settings**.

   The **Third Party Credentials** page opens.
3. Click the **Pulse** tab.
4. Click **Enable Pulse**.

   The system displays the **Enable Pulse** window.
5. Click **Enable Pulse**.

   The system enables Pulse and displays the message Pulse setting has been enabled successfully.

Disabling Nutanix Enterprise AI Pulse Telemetry

Pulse shares data about your Nutanix Enterprise AI application usage with Nutanix. This topic describes the steps to disable Nutanix Enterprise AI Pulse telemetry.

About this task

To disable Nutanix Enterprise AI Pulse telemetry, follow these steps:

> **Caution:** Nutanix recommends that you enable Pulse to allow Nutanix Support to receive cluster data and deliver proactive and context-aware support.

Procedure

1. Log in to Nutanix Enterprise AI as an AI/ML Admin user.
2. From the left navigation bar, click **Settings**.

   The **Third Party Credentials** page opens.
3. Click the **Pulse** tab.
4. Click **Disable Pulse**.

   The system displays the **Disable Pulse** window.
5. Click **Disable Pulse**.

   The system enables Pulse and displays the message Pulse setting has been disabled successfully.

CONFIGURING OPENTELEMETRY COLLECTOR TO VIEW NUTANIX ENTERPRISE AI METRICS

Configure OpenTelemetry Collector to export the metrics available in Nutanix Enterprise AI and view the metrics in any OpenTelemetry compliant observability tool.

Before you begin

- Ensure that you install an active version of OpenTelemetry Collector. For more information, see OpenTelemetry documentation.
- Ensure that you enable the appropriate access and permissions in OpenTelemetry Collector. For more information, see OpenTelemetry documentation.
- Ensure that each receiver has the correct namespace and selector labels configured in *kubernetes_sd_configs* for your cluster.

About this task

The Nutanix Enterprise AI dashboard displays inference request metrics, such as Request Count By Status, Request Latency, and infrastructure metrics such as CPU, Memory, and GPU utilization.

In addition, Nutanix Enterprise AI provides other metrics, categorised by the following metric sources:

- API server metrics such as Inference Request Count, Latency, and Throughput.
- GPU metrics exposed by NVIDIA DCGM Exporter
- Node metrics exposed by Node Exporter
- vLLM endpoint metrics exposed by vLLM Runtime
- TGI endpoint metrics exposed by TGI Engine
- NIM endpoint metrics exposed by NIM Engine

To collect these metrics, add Prometheus receivers to the OpenTelemetry Collector. You can then export these metrics to any OpenTelemetry compatible observability tool to view these metrics.

The configuration in this procedure creates Prometheus receivers for each metric source in Nutanix Enterprise AI. If you have an OpenTelemetry Collector installed, add these Prometheus receivers to your configuration to start collecting metrics from all Nutanix Enterprise AI sources.

To configure OpenTelemetry Collector to export Nutanix Enterprise AI metrics, follow these steps:

Procedure

1. Access the configuration file of the OpenTelemetry Collector installed on your cluster.
2. Enter the following Prometheus receiver configuration in the configuration file.

   Note: Ensure that the *kubernetes_sd_configs* of the scrape jobs in the configuration file have the correct namespace and selector labels configured according to your cluster.

```
receivers:
  prometheus:
    config:
```

```
    scrape_configs:
      - job_name: nai
        scrape_interval: 10s
        metrics_path: /metrics/all
        static_configs:
          - targets:
              - nai-api.nai-system.svc.cluster.local:8080
      - job_name: dcgm-exporter
        scrape_interval: 10s
        metrics_path: /metrics
        relabel_configs:
          - source_labels:
              [
                __meta_kubernetes_pod_container_port_number,
              ]
            action: keep
            regex: 9400
        kubernetes_sd_configs:
          - role: pod
            namespaces:
              names:
                - <GPU_OPERATOR_NAMESPACE>
            selectors:
              - role: pod
                label: "app=nvidia-dcgm-exporter"
      - job_name: node-exporter
        scrape_interval: 10s
        metrics_path: /metrics
        relabel_configs:
          - source_labels:
              [
                __meta_kubernetes_pod_container_port_number,
              ]
            action: keep
            regex: 9100
        kubernetes_sd_configs:
          - role: pod
            namespaces:
              names:
                - <NODE_EXPORTER_NAMESPACE>
            selectors:
              - role: pod
                label: "app.kubernetes.io/name=prometheus-node-exporter"
      - job_name: vllm-cpu-endpoints
        scrape_interval: 10s
        metrics_path: /metrics
        kubernetes_sd_configs:
          - role: pod
            namespaces:
              names:
                - nai-admin
            selectors:
              - role: pod
                label: "endpoint.iep.nai.nutanix.com/engine=vllm-cpu"
        relabel_configs:
          - source_labels:
              [
                __meta_kubernetes_pod_container_port_number,
              ]
            action: keep
            regex: 8080
      - job_name: vllm-endpoints
```

```
          scrape_interval: 10s
          metrics_path: /metrics
          kubernetes_sd_configs:
            - role: pod
              namespaces:
                names:
                  - nai-admin
              selectors:
                - role: pod
                  label: "endpoint.iep.nai.nutanix.com/engine=vllm"
          relabel_configs:
            - source_labels:
                [
                  __meta_kubernetes_pod_container_port_number,
                ]
              action: keep
              regex: 8080
        - job_name: tgi-endpoints
          scrape_interval: 10s
          metrics_path: /metrics
          kubernetes_sd_configs:
            - role: pod
              namespaces:
                names:
                  - nai-admin
              selectors:
                - role: pod
                  label: "endpoint.iep.nai.nutanix.com/engine=tgi"
          relabel_configs:
            - source_labels:
                [
                  __meta_kubernetes_pod_container_port_number,
                ]
              action: keep
              regex: 8080
            - source_labels: [__meta_kubernetes_namespace]
              target_label: namespace
            - source_labels: [__meta_kubernetes_pod_name]
              target_label: pod
        - job_name: nim-endpoints
          scrape_interval: 10s
          metrics_path: /metrics
          kubernetes_sd_configs:
            - role: pod
              namespaces:
                names:
                  - nai-admin
              selectors:
                - role: pod
                  label: "endpoint.iep.nai.nutanix.com/engine=nim"
          relabel_configs:
            - source_labels:
                [
                  __meta_kubernetes_pod_container_port_number,
                ]
              action: keep
              regex: 8000
```

3. Add the Prometheus receiver in the metrics pipeline definition.

```
service:
     pipelines:
```

```
        metrics:
          receivers: [prometheus]
```

4. Save the file.

ACCESSING ONLINE HELP

NAI includes online help documentation that you can access at any time.

About this task

To access the online help documentation, follow these steps.

Procedure

1. Log in to NAI.
2. Click the user name in the upper-right corner.
3. From the dropdown menu, select **Online Documentation**.
   The online help documentation appears in a new tab or window. These pages are located on the Nutanix support portal.

NUTANIX ENTERPRISE AI SUPPORT BUNDLE

The Nutanix Enterprise AI Support Bundle contains all the necessary Kubernetes pod logs and resource information required to debug an ongoing issue for your NAI deployment. This will assist the Nutanix support team with troubleshooting.

The procedures to generate the support bundle are as follows:

- If you have installed Nutanix Kubernetes Platform (NKP), see Generating Support Bundle in an NKP Environment on page 102.
- If you have not installed Nutanix Kubernetes Platform (NKP), see Generating Support Bundle in a Non-NKP Environment on page 102.

Generating Support Bundle in an NKP Environment

Generate support bundle in a NKP environment.

About this task

To generate support bundle in a NKP environment, follow these steps:

Procedure

1. Download NKP CLI.

   For more information, see Downloading NKP.

2. Collect the logs, metrics, and configuration data from your Kubernetes cluster in a zipped support bundle:

   ```
   nkp diagnose --kubeconfig=cluster-kubeconfig.yaml
   ```

   For more information on the NKP diagnose command, see NKP Troubleshooting Guide.

   The zipped support bundle is downloaded.

3. Extract the zipped support bundle.
   The downloaded artifacts are displayed.

What to do next

Share the generated bundle with the Nutanix Support Team.

For more information, see KB-1294 Uploading Files for Nutanix Support Using FTP, SFTP or the Customer Portal.

Generating Support Bundle in a Non-NKP Environment

Generate support bundles in a non-NKP environment.

Before you begin

Ensure that you have a Kubernetes cluster and local `kubectl` access to the cluster. Install `kubectl` from kubectl.

About this task

To generate support bundles in a non-NKP environment, follow these steps:

Procedure

1. Download the binary for your operating system and architecture from the GitHub releases:

    » For Linux AMD64:

    ```
    curl -L https://github.com/replicatedhq/troubleshoot/releases/download/v0.119.0/
    support-bundle_linux_amd64.tar.gz | tar xzvf -
    ```

    » For Linux ARM64:

    ```
    curl -L https://github.com/replicatedhq/troubleshoot/releases/download/v0.119.0/
    support-bundle_linux_arm64.tar.gz | tar xzvf -
    ```

    » For macOS ARM64 (Apple Silicon):

    ```
    curl -L https://github.com/replicatedhq/troubleshoot/releases/download/v0.119.0/
    support-bundle_darwin_arm64.tar.gz | tar xzvf -
    ```

    For any other OS, architecture, please refer to this release page.

2. Verify if the plugin is installed correctly:

    `./support-bundle version`

3. Set Up Kubernetes Configuration. Ensure your cluster's KUBECONFIG is properly configured:

    `export KUBECONFIG=/path/to/your/kubeconfig`

4. Collect the support bundle:

    Run the support bundle collection with your collector configuration:

    ```
    ./support-bundle support-bundle-collector.yaml
    ```

    This will create a zipped archive containing logs, metrics, and configuration data from your Kubernetes cluster.

    The format of the `support-bundle-collector.yaml` file is as follows:

    ```
    apiVersion: troubleshoot.sh/v1beta2
    kind: SupportBundle
    metadata:
      name: nai-support-bundle
    spec:
      collectors:
        - helm: {} # collects information about all helm releases in all namespaces
        - nodeMetrics: {} # collects metrics for all the nodes, all containers within the
     nodes
        - logs:
            collectorName: "nai-system-logs"
            name: "nai-system-logs"
            namespace: nai-system # collects logs for all the containers in nai-system
     namespace
        - logs:
            collectorName: "nai-admin-logs"
            name: "nai-admin-logs"
            namespace: nai-admin
    ```

    The zipped support bundle is downloaded.

5. Extract the zipped support bundle.
    The downloaded artifacts are displayed.

6. Ensure the support bundle doesn't contain any sensitive information.

    You can redact sensitive information by using the https://troubleshoot.sh/docs/redact/ feature.

What to do next

Share the generated bundle with the Nutanix Support Team.

For more information, see KB-1294 Uploading Files for Nutanix Support Using FTP, SFTP or the Customer Portal.

GLOSSARY

For terms used in this guide, see Nutanix Glossary.

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